Supplementary Material to "Information Lower Bounds of Trajectory Health Indicators for Remaining Useful Life Prediction"

This Supplementary Material contains the proofs and supplementary results, the details of the real-data tests, the details of the benchmark (learned encoders, data, protocols, detailed tables, initial protocol) and the figures and tables moved out of the paper. Sections, theorems, tables and references of the paper are cited by their numbers in the paper; the bibliography of this document lists the references that it cites.

S1. Proofs and supplementary results

S1.1. Proofs of the results of Sections 2 to 4

Proof of Theorem 1. (a) Let $E = T_t - g(H_{1:t})$. Differential entropy is translation invariant [5, Sect. 8.6], so $h(E \mid H_{1:t}) = h(T_t \mid H_{1:t})$ ($g(H_{1:t})$ is constant given $H_{1:t}$). Conditioning does not increase entropy [5, Sect. 2.6], so $h(E \mid H_{1:t}) \leq h(E)$, and the Gaussian maximises entropy at fixed second moment [5, Sect. 8.6], so $h(E) \leq \frac{1}{2}\ln(2\pi e\,\mathbb{E}[E^2])$. Combining the three relations gives $\mathbb{E}[E^2] \geq \frac{1}{2\pi e} e^{2h(T_t|H_{1:t})}$. The identity $h(T_t \mid H_{1:t}) = h(T_t) - I(T_t; H_{1:t})$ (definition of mutual information) gives the second form. By (1), $h(T_t \mid H_{1:t}) \geq h(T_t \mid \mathbf{X}_{1:t})$, and the ratio follows from $h(T_t \mid H_{1:t}) - h(T_t \mid \mathbf{X}_{1:t}) = I(T_t; \mathbf{X}_{1:t}) - I(T_t; H_{1:t})$. Applying the same steps conditionally on $H_{1:t} = \mathsf{h}$ and then averaging gives the sharper *pointwise* form $\varepsilon^2 \geq \mathbb{E}_H\big[e^{2h(T_t|H_{1:t}=\mathsf{h})}\big]/(2\pi e) \geq \mathcal{B}_{\text{info}}(H_{1:t})$ (Jensen's inequality for the convex exponential); it is the form reported in the simulations of Sections 3 and 4.1. (b) Rearrange the first form of (a). (c) Apply (a) conditionally on t and average. The chain rule of mutual information [5, Sect. 2.5] gives the decomposition. For the inequality, (b) conditionally on t yields $I(T_t; H_{1:t} \mid t = \tau) \geq h(T_t \mid t = \tau) - \frac{1}{2}\ln(2\pi e\,\varepsilon_\tau^2)$ with ε_τ^2 the conditional MSE; averaging over τ and using Jensen's inequality for the concave logarithm [5, Sect. 2.6], $\mathbb{E}_\tau \ln \varepsilon_\tau^2 \leq \ln \mathbb{E}_\tau \varepsilon_\tau^2 = \ln \varepsilon^2$, gives the stated bound. □

Proof of Theorem 2. Since $\mathbb{E}[T_t - m(\mathbf{z}) \mid \mathbf{z}, H_{1:t}] = 0$ and T_t is conditionally independent of $H_{1:t}$ given $\mathbf{z}$, the cross term vanishes and $\mathbb{E}[(T_t - \hat{T}_t)^2] = \mathbb{E}[\mathrm{Var}(T_t \mid \mathbf{z})] + \mathbb{E}[(m(\mathbf{z}) - \hat{T}_t)^2]$, the usual split of the squared error into an irreducible and an estimation part [7, Sect. 1.5.5]. The first term is $\mathbb{E}[a\sigma_B^2/\mu^3]$. For the second, apply Theorem 1 conditionally on $H_{1:t} = \mathsf{h}$ to the target $m(\mathbf{z})$, which gives $\mathbb{E}[(m - \hat{T}_t)^2 \mid H = \mathsf{h}] \geq e^{2h(m|H=\mathsf{h})}/(2\pi e)$, and average over h. For the second inequality, the exponential is convex, so (Jensen [5, Sect. 2.6]) $\mathbb{E}_H[e^{2h(m|H=\mathsf{h})}] \geq e^{2h(m|H)}$ with $h(m \mid H) = \mathbb{E}_H h(m \mid H = \mathsf{h})$; and $m \to \mathbf{X}_{1:t} \to H_{1:t}$ is Markov, so $h(m \mid H) = h(m) - I(m; H) = h(m \mid \mathbf{X}) + I(m; \mathbf{X}) - I(m; H) = h(m \mid \mathbf{X}) + (1 - \eta_m) I(m; \mathbf{X})$. □

Proof of Theorem 4. (a) Because $\mathbf{z} = \hat{\mathbf{z}} + \boldsymbol{\varepsilon}$ with $\boldsymbol{\varepsilon} \sim \mathcal{N}(0, \mathbf{P})$ independent of $\hat{\mathbf{z}}$, we have $\mathrm{Cov}(\mathbf{z}, \hat{\mathbf{z}}) = \boldsymbol{\Lambda}$ and $\mathrm{Var}(\mathbf{g}^\top \mathbf{z}) = \mathbf{g}^\top \boldsymbol{\Sigma} \mathbf{g} = \mathbf{g}^\top \boldsymbol{\Lambda} \mathbf{g} + \mathbf{g}^\top \mathbf{P} \mathbf{g}$. The pair $(\mathbf{g}^\top \mathbf{z}, h_{\mathbf{u}})$ is jointly Gaussian with $\mathrm{Cov} = \mathbf{g}^\top \boldsymbol{\Lambda} \mathbf{u}$ and $\mathrm{Var}(h_{\mathbf{u}}) = \mathbf{u}^\top \boldsymbol{\Lambda} \mathbf{u}$, so the Gaussian conditioning formula [7, Sect. 2.3] gives $\mathrm{Var}(m_L \mid h_{\mathbf{u}}) = \mathbf{g}^\top \boldsymbol{\Lambda} \mathbf{g} + \mathbf{g}^\top \mathbf{P} \mathbf{g} - (\mathbf{g}^\top \boldsymbol{\Lambda} \mathbf{u})^2/(\mathbf{u}^\top \boldsymbol{\Lambda} \mathbf{u}) = \mathbf{g}^\top \mathbf{P} \mathbf{g} + \mathbf{g}^\top \boldsymbol{\Lambda} \mathbf{g}\,(1 - \cos^2 \psi)$.
(b) In the linearised model the conditional distribution of m_L is Gaussian, so the entropy power equals the variance. Taking the ratio of (a) to the case $h = \hat{\mathbf{z}}$, for which $\mathrm{Var}(m_L \mid \hat{\mathbf{z}}) = \mathbf{g}^\top \mathbf{P} \mathbf{g}$, gives the first formula. Moreover, by the formula for the mutual information of jointly Gaussian variables [5, Chap. 8], $I(m_L; \mathbf{X}) = \frac{1}{2}\ln(\mathrm{Var}(m_L)/\mathbf{g}^\top \mathbf{P} \mathbf{g}) = \frac{1}{2}\ln(1 + \rho)$ and $I(m_L; h_{\mathbf{u}}) = \frac{1}{2}\ln\big((1 + \rho)/(1 + \rho \sin^2 \psi)\big)$; their ratio is η_m.

(c) $\sin^2\psi = 0$ iff $\mathbf{\Lambda}^{1/2}\mathbf{g} \parallel \mathbf{\Lambda}^{1/2}\mathbf{u}$ iff $\mathbf{u} \parallel \mathbf{g}$ since $\mathbf{\Lambda} \succ 0$. For $\mathbf{u} = \mathbf{g}$ the loss vanishes, and Lemma 1 shows that no scalar linear functional of $\mathbf{X}$ can do better. □

Proof of Proposition 1. Because D and r are independent, $\mathrm{Cov}(D, x) = \bar{c}\, v$, and $\mathrm{Var}(x) = \mathrm{Var}(c_r D) + \mathrm{Var}(o_r) + 2\bar{d}\,\mathrm{Cov}(c_r, o_r) + \sigma^2 = V_x$, since $\mathrm{Cov}(c_r D, o_r) = \bar{d}\,\mathrm{Cov}(c_r, o_r)$ and $\mathrm{Var}(c_r D) = \mathbb{E}[c_r^2]\mathbb{E}[D^2] - (\bar{c}\bar{d})^2$; the best linear predictor of D from x has error $v - \mathrm{Cov}(D, x)^2/\mathrm{Var}(x)$ (the conditional variance of a Gaussian pair with the same second moments [7, Sect. 2.3]). For the limit use $\mathbb{E}[c_r^2]\mathbb{E}[D^2] - (\bar{c}\bar{d})^2 = \bar{c}^2 v + \mathrm{Var}(c_r)(v + \bar{d}^2)$ and $\mathrm{Var}(c_r)\bar{d}^2 + \mathrm{Var}(o_r) + 2\bar{d}\mathrm{Cov}(c_r, o_r) = \mathrm{Var}(\bar{d}c_r + o_r)$, so that at $\sigma = 0$, $V_x = \bar{c}^2 v + \mathrm{Var}(c_r)v + \mathrm{Var}(\bar{d}c_r + o_r)$. Given the regime, the estimator $(x - o_r)/c_r$ is unbiased with variance σ^2/c_r^2. □

Proof of Proposition 2. The level error obeys $\varepsilon_k = (1-\omega)(\varepsilon_{k-1} + e_k) + \omega n_k$ with $\varepsilon_0 = 0$ (substitute $H_k = D_k + \varepsilon_k$, $u_k = v_k + e_k$ and $l_k = D_k + n_k$ into the update, with $D_k = D_{k-1} + v_k$ the true level and v_k its true increment); write $\alpha = 1 - \omega$. For $\omega = 0$, $\varepsilon_t = \sum_{k \le t} e_k$ has mean tb and variance $t\sigma_w^2$. For $\omega > 0$ the mean $\bar{\varepsilon}_k$ satisfies $\bar{\varepsilon}_k = \alpha(\bar{\varepsilon}_{k-1} + b)$, with fixed point $\alpha b/(1-\alpha) = (1-\omega)b/\omega$ approached geometrically, and the variance $s_k^2 = \alpha^2 s_{k-1}^2 + \alpha^2\sigma_w^2 + \omega^2\sigma_n^2$ (independent noise terms) has fixed point $(\alpha^2\sigma_w^2 + \omega^2\sigma_n^2)/(1-\alpha^2)$ with $1 - \alpha^2 = \omega(2-\omega)$. □

S1.2. Relation to the Bayesian Cramér–Rao bound

Remark 3 (Relation to the Bayesian Cramér–Rao bound)**.** *Let $J(\mathsf{h})$ be the Fisher information, with respect to location, of the posterior density $p(T_t \mid H_{1:t} = \mathsf{h})$. Under the regularity conditions of the van Trees inequality [1, 2] every estimator satisfies $\mathbb{E}[(T_t - \hat{T}_t)^2] \ge 1/\mathbb{E}_H[J(H)]$. Stam's inequality [3] states that $e^{2h(T_t|H=\mathsf{h})}/(2\pi e) \ge 1/J(\mathsf{h})$ for every h. Taking the geometric mean over h and using that the geometric mean of $1/J$ is at least the reciprocal of the arithmetic mean of J,*

$$\mathcal{B}_{\mathrm{info}}(H_{1:t}) = \exp\Big(\mathbb{E}_H \ln \frac{e^{2h(T_t|H=\mathsf{h})}}{2\pi e}\Big) \;\ge\; \exp\Big(\mathbb{E}_H \ln \frac{1}{J(H)}\Big) \;\ge\; \frac{1}{\mathbb{E}_H[J(H)]}, \tag{1}$$

so, wherever the Bayesian Cramér–Rao bound applies, the entropy-power bound is not weaker, and it requires no differentiability of the densities. For posterior filtering the Fisher-information route is developed in [4].

S1.3. State–kinematic decomposition and pointwise shrinkage

Proposition 3 (State–kinematic decomposition)**.** *Let $\Delta H_k = H_k - H_{k-1}$ denote the increments of the HI. The map $H_{1:t} \mapsto (H_t, \Delta H_{2:t})$ is a linear bijection, hence*

$$I(T_t; H_{1:t}) = \underbrace{I(T_t; H_t)}_{\mathcal{I}_{\mathrm{state}}(t)} + \underbrace{I(T_t; \Delta H_{2:t} \mid H_t)}_{\mathcal{I}_{\mathrm{kin}}(t)}. \tag{2}$$

Proof. The map has a unit-determinant matrix, so $h(\mathbf{AY}) = h(\mathbf{Y}) + \ln|\det \mathbf{A}|$ [5, Sect. 8.6] and mutual information is unchanged ($H_k = H_t - \sum_{j=k+1}^{t} \Delta H_j$ gives the inverse). The identity is then the chain rule of mutual information [5, Sect. 2.5]. □

Remark 4. *The decomposition is an identity; its content lies in which operations remove which part. If the HI is the noise-free level of a process with constant drift started at zero, H_t/t is a sufficient statistic for the drift, and $\mathcal{I}_{\mathrm{kin}}$ about the drift is zero. Kinematic information is non-trivial when the velocity varies in time, for example under a state-dependent drift or a random-walk velocity, as in late-life acceleration.*

Proposition 4 (Pointwise shrinkage)**.** *Let the shrinkage of Theorem 3 act pointwise in time, $\tilde{H}_k = \phi_k(H_k, \xi_k)$. Then*

$$\ln \frac{\mathcal{B}_{\text{info}}(\tilde{H}_{1:t})}{\mathcal{B}_{\text{info}}(H_{1:t})} = 2\big(\Delta\mathcal{I}_{\text{state}} + \Delta\mathcal{I}_{\text{kin}}\big), \qquad (3)$$
$$\Delta\mathcal{I}_{\text{state}} := \mathcal{I}_{\text{state}}(H) - \mathcal{I}_{\text{state}}(\tilde{H}) \geq 0, \quad \Delta\mathcal{I}_{\text{kin}} := \mathcal{I}_{\text{kin}}(H) - \mathcal{I}_{\text{kin}}(\tilde{H}),$$

where $\Delta\mathcal{I}_{\text{kin}}$ has no definite sign in general, so that kinematic information alone does not determine the increase of the bound. In the Gaussian shrinkage of Theorem 3(b), write $\mathcal{I}^{\vartheta}_{\text{state}}(\gamma) = I(\vartheta; \tilde{H}^{\gamma}_t)$ and $\mathcal{I}^{\vartheta}_{\text{kin}}(\gamma) = I(\vartheta; \Delta\tilde{H}^{\gamma}_{2:t} \mid \tilde{H}^{\gamma}_t)$; then $\mathcal{I}^{\vartheta}_{\text{state}}$ is non-increasing in γ and $0 \leq \mathcal{I}^{\vartheta}_{\text{state}}, \mathcal{I}^{\vartheta}_{\text{kin}} \leq \mathcal{I}^{\vartheta}_{\text{tot}}(\gamma) \to 0$, although $\mathcal{I}^{\vartheta}_{\text{kin}}$ need not be monotone.

Proof. Apply Proposition 3 to H and to $\tilde{H}$ and take the difference in Theorem 3(a). Because $\tilde{H}_t = \phi_t(H_t, \xi_t)$ is a randomised function of H_t alone, the DPI gives $I(T_t; \tilde{H}_t) \leq I(T_t; H_t)$, that is, $\Delta\mathcal{I}_{\text{state}} \geq 0$. For non-pointwise operators (for example smoothing over history) even this sign can fail. In the Gaussian shrinkage, the argument of the proof of Theorem 3(b) applied to the scalar observation at time t gives the monotonicity of $\mathcal{I}^{\vartheta}_{\text{state}}$; since $\mathcal{I}^{\vartheta}_{\text{state}} = I(\vartheta; \tilde{H}_t) \leq I(\vartheta; \tilde{H}_{1:t})$ and $\mathcal{I}^{\vartheta}_{\text{kin}} = \mathcal{I}^{\vartheta}_{\text{tot}} - \mathcal{I}^{\vartheta}_{\text{state}} \leq \mathcal{I}^{\vartheta}_{\text{tot}}$, both are dominated by $\mathcal{I}^{\vartheta}_{\text{tot}}$, which vanishes as $\gamma \to 1$. □

S1.4. Proof of Theorem 3

Proof. (a) $\tilde{H}$ is a randomised function of H only, so the Markov property holds and the DPI gives the first claim; the ratio follows from the definition of $\mathcal{B}_{\text{info}}$. (b) With $\lambda = 1 - \gamma > 0$, $\tilde{H}^{\gamma}$ is an invertible affine image of $(H - m) + \xi^{(2)}/\lambda$, that is, of H observed through additive Gaussian noise of variance σ_2^2/λ^2. For $\gamma_2 > \gamma_1$ the noise of the second channel is the first plus independent Gaussian noise, so the DPI [5, Sect. 2.8] gives monotonicity of $\mathcal{I}^{\vartheta}_{\text{tot}}$. As $\lambda \to 0$ the noise variance diverges; in a linear-Gaussian channel $I(\vartheta; Y) = \frac{1}{2}\ln\big(\operatorname{Var}(\vartheta)/\operatorname{Var}(\vartheta \mid Y)\big)$ [5, Chap. 9] and $\operatorname{Var}(\vartheta \mid Y) \to \operatorname{Var}(\vartheta)$, so the information vanishes. For the limit of Mon, as $\lambda \to 0$ at fixed t the increments converge to $\Delta m + \xi^{(2)}_{k+1} - \xi^{(2)}_k$, a stationary 1-dependent sequence. The indicators of a positive increment at odd k involve disjoint pairs of the $\xi^{(2)}$ and are therefore i.i.d., as are those at even k, so the strong law of large numbers [6, Chap. 2] applies to each subsequence and the fraction of positive increments tends to $p = \Phi_{\mathcal{N}}(\Delta m/(\sqrt{2}\sigma_2))$ and, letting $t \to \infty$, Mon $\to |2p - 1|$. □

S1.5. Non-Gaussianity gap

Proposition 5 (Non-Gaussianity gap)**.** *Let $\hat{m}_H = \mathbb{E}[m \mid H]$ be the Bayes-optimal estimator of m and let $D_{\mathsf{h}} := D_{\text{KL}}\big(p(m \mid H = \mathsf{h}) \,\|\, \mathcal{N}(\mathbb{E}[m \mid H = \mathsf{h}], \operatorname{Var}(m \mid H = \mathsf{h}))\big) \geq 0$ be the Kullback–Leibler (KL) divergence of the posterior from the Gaussian with the same mean and variance. Then*

$$\mathbb{E}\big[(m - \hat{m}_H)^2\big] - \mathbb{E}_H\left[\frac{e^{2h(m|H=\mathsf{h})}}{2\pi e}\right] = \mathbb{E}_H\Big[\operatorname{Var}(m \mid H)\big(1 - e^{-2D_H}\big)\Big] \geq 0, \qquad (4)$$

with equality if and only if $p(m \mid H)$ is Gaussian almost surely.

Proof. For a density p with mean μ_p and variance σ_p^2 and the Gaussian $q = \mathcal{N}(\mu_p, \sigma_p^2)$, $\int p \ln q = -\frac{1}{2}\ln(2\pi\sigma_p^2) - \frac{1}{2}$ because q depends on x only through the first two moments of p. Hence $D_{\text{KL}}(p\|q) = -h(p) + \frac{1}{2}\ln(2\pi e\sigma_p^2)$ (the identity behind the maximum-entropy

property of the Gaussian [5, Sect. 8.6]), that is, $e^{2h(p)}/(2\pi e) = \sigma_p^2 e^{-2D_{\mathrm{KL}}(p\|q)}$. Apply this to $p = p(m \mid H = \mathsf{h})$, use $\mathbb{E}[(m - \hat{m}_H)^2 \mid H = \mathsf{h}] = \mathrm{Var}(m \mid H = \mathsf{h})$, which holds because the posterior mean is the Bayes-optimal estimator under squared loss [7, Sect. 1.5.5], and average over h. The right-hand side vanishes if and only if $D_H = 0$ almost surely, that is, if and only if the posterior is Gaussian. □

S1.6. Reduction to functionals of the sufficient statistic

Lemma 1 (Reduction to functionals of $\hat{\mathbf{z}}$)**.** *For every scalar linear functional $w^\top \mathbf{X}_{1:t}$ there is* $\mathbf{u}$ *such that* $\mathrm{corr}(m_L, w^\top \mathbf{X})^2 \le \mathrm{corr}(m_L, \mathbf{u}^\top \hat{\mathbf{z}})^2$*. Hence no scalar linear HI is more informative about m_L than the best $h_{\mathbf{u}}$, because in a Gaussian model $I = -\frac{1}{2}\ln(1 - \mathrm{corr}^2)$. Here* corr *denotes the correlation coefficient.*

Proof. Let $\mathbf{C} = \mathrm{Cov}(\mathbf{z}, \mathbf{X})$ and $\mathbf{V} = \mathrm{Var}(\mathbf{X})$, so $\hat{\mathbf{z}} = \mathbf{z}_0 + \mathbf{C}\mathbf{V}^{-1}(\mathbf{X} - \mathbb{E}\mathbf{X})$ and $\mathbf{\Lambda} = \mathbf{C}\mathbf{V}^{-1}\mathbf{C}^\top$. Put $\mathbf{u} = \mathbf{\Lambda}^{-1}\mathbf{C}w$. Then $\mathrm{Cov}(\hat{\mathbf{z}}, w^\top \mathbf{X}) = \mathbf{C}\mathbf{V}^{-1}\mathbf{V}w = \mathbf{C}w = \mathbf{\Lambda}\mathbf{u}$, so $w^\top \mathbf{X} = \mathbf{u}^\top \hat{\mathbf{z}} + e$ with e uncorrelated with $\hat{\mathbf{z}}$ and, since $\mathrm{Cov}(\mathbf{z}, w^\top \mathbf{X}) = \mathbf{C}w = \mathbf{\Lambda}\mathbf{u} = \mathrm{Cov}(\mathbf{z}, \mathbf{u}^\top \hat{\mathbf{z}})$, also uncorrelated with $\mathbf{z}$. Therefore $\mathrm{Cov}(m_L, w^\top \mathbf{X}) = \mathrm{Cov}(m_L, \mathbf{u}^\top \hat{\mathbf{z}}) = \mathbf{g}^\top \mathbf{\Lambda}\mathbf{u}$ while $\mathrm{Var}(w^\top \mathbf{X}) = \mathbf{u}^\top \mathbf{\Lambda}\mathbf{u} + \mathrm{Var}(e) \ge \mathrm{Var}(\mathbf{u}^\top \hat{\mathbf{z}})$, which gives the claim. □

S1.7. Proof of Corollary 1

Proof. (a) By Theorem 4, the gradient of the frozen-velocity RUL at the operating point is $\mathbf{g} = -\bar{V}^{-1}(1, \bar{a}/\bar{V})^\top$, which is proportional to $(1, \bar{\tau})^\top$; the Euclidean angle of this direction from the level axis is $\arctan \bar{\tau}$. (b) By Theorem 4(c), the increase of the bound at stage j vanishes only for $\mathbf{u} \parallel \mathbf{g}_j$, so it vanishes at both stages only if $\mathbf{g}_1 \parallel \mathbf{g}_2$. The map $\mathbf{u} \mapsto \max_j \rho_j \sin^2 \psi_j(\mathbf{u})$ is continuous and positive on the unit circle, hence has a positive minimum δ. For the quantitative statement note that the $\mathbf{\Lambda}$-angle between lines satisfies the triangle inequality $\psi_{12} \le \psi_1 + \psi_2$. Indeed, for unit vectors a, b in the $\mathbf{\Lambda}$-inner product the angle $\arccos\langle a, b\rangle$ is the geodesic distance on the unit sphere, which is a metric, and the angle between the lines through a and b is $\min_{s=\pm 1} \arccos\langle a, sb\rangle$. Choose the sign of the representative of $\mathbf{u}$ so that its angle to the representative of $\mathbf{g}_1$ is ψ_1, then the sign of the representative of $\mathbf{g}_2$ so that its angle to that of $\mathbf{u}$ is ψ_2; the spherical triangle inequality bounds the angle between the representatives of $\mathbf{g}_1$ and $\mathbf{g}_2$ by $\psi_1 + \psi_2$, and ψ_{12} is not larger. Therefore $\max_j \psi_j \ge \psi_{12}/2$ and $\sin^2$ is increasing on $[0, \pi/2]$. The last sentence is Theorem 4(c) applied at each stage. □

S2. Details of the real-data tests

Turbofan engines: set-up. Each subset uses its training engines (run to failure); the 14 informative sensors are standardised with the training engines of the fold, within regime for FD002 and FD004 (six regimes from k-means clustering of the operating settings). The ridge HI is a ridge regression (penalty 10) of the life fraction on the standardised sensors, smoothed by a trailing mean of 5 cycles and mapped affinely to zero at the start and one at the end of life with the training engines of the fold; PCA-HI is the first principal component. The trajectory summary of an HI consists of its trailing mean over 10 cycles (level), its least-squares slope over the last 30 cycles (velocity), its values 20 and 50 cycles earlier, and the age; the raw-sensor predictor uses level and slope of every sensor and the age. Predictors are histogram-based gradient-boosted regression trees (200 iterations, learning rate 0.08, 31 leaves, at least 40 samples per leaf, L_2 penalty 1) trained on all pairs of the training engines of a fold. The lifetime distribution is estimated by

a Gaussian kernel density of $\ln L$ (Scott bandwidth) of the training engines of the fold; the conditional density of the RUL at age t is $f_L(t+\tau)/S_L(t)$ and $h(T_t \mid t)$ is evaluated numerically on a grid of 0.5 cycles. Bootstrap intervals resample engines (300 replicates).

Lithium-ion cells. Each cycle of the processed records yields the discharge capacity, the charge duration (samples with current above 0.01 A) and the charge voltage when the charged capacity first reaches 27% and 73% of the nominal capacity; the HIs use the last three quantities. The charge duration is divided by its mean over cycles 2 to 11 of the same cell and the voltages are reduced by their means over these cycles, and the result is standardised with the training cells of the fold. Time scales are set by the lifetimes. For HUST the minimum age is 100 cycles, ages are sampled every 5 cycles, the windows are 20 cycles (level) and 100 cycles (velocity), the lags are 100 and 250 cycles and the grid of the conditional entropy has a step of 2 cycles. For MATR the corresponding values are 50, 3, 10, 50, 50 and 125 cycles and a step of 1 cycle. The trees have at most 15 leaves, and all other settings are those of the engines. The capacity reference uses level and velocity of the relative discharge capacity and the age. The end of life is the first cycle at which the running median of the discharge capacity over 5 cycles falls below 80% of the mean over the first 5 cycles (HUST), or the life label of the source, at which the recordings end (MATR; recordings that end more than 10 cycles before the label would be excluded, and none was). For Tongji the minimum age is 30 cycles, ages are sampled every 2 cycles, the windows are 10 and 40 cycles, the lags 40 and 100 cycles and the step 1 cycle, and the end of life is the first cycle at which the running median of the discharge capacity falls below 80% of the nominal capacity; cells that do not reach it, or reach it within 100 cycles, are excluded. The capacity levels of the charge-voltage and discharge-voltage features are 3/11 and 8/11 of the nominal capacity. The processed records are unpickled with a restricted reader that admits only array types. The protocol analysis of Section 5 uses the three charging features standardised over the population (blind) or relative to the own cycles 2 to 11 (normalised), with the ridge regression of the engines.

Additional analyses. Candidate HIs. For each data set, 40 weight vectors with independent standard normal entries (signed so that the HI increases with age on the training units) are applied to the normalised sensors (engines) or charging features (cells), together with the PCA-HI and the ridge HI; each HI is smoothed over 5 cycles and mapped affinely to zero at the start and one at the end of life with the training units of the fold. The information lower bound is computed as in the main analysis (ages sampled every 3 cycles for engines), and the geometric score is the sum of the monotonicity, the trendability (absolute correlation with time, averaged over units) and the prognosability on the held-out units; the Spearman correlation is taken over the 42 HIs. *Decision evaluation.* The predictions of the age-only baseline, PCA-HI, Geo-opt HI, ridge HI and the ten-seed ensembles of PGIB and PGIB-KF on the official test engines are used; the margin added to the predicted RUL is the 5% or 10% quantile of the residuals (true minus predicted RUL) of the method, estimated on four of five folds of engines and applied to the fifth (20 repetitions), so that about this fraction of the engines fails; intervals are bootstrap intervals over engines (400 replicates). *End-to-end predictor.* Gradient boosting with the settings of the engines on the regime-normalised level and slope (windows of 10 and 30 cycles) of the 14 sensors and the age, trained on all cycles of all training engines and evaluated at the last cycle of each test engine; the information lower bound uses the lifetime distribution of the training engines. *Inclusion rule.* For Tongji, cells that never fall below

Table S1: Tongji: main analysis (88 cells) and sensitivity to the inclusion rule (119 cells, cells that never reached the threshold counted as failed at their last recorded cycle).

	Main (88 cells)	All cells (119 cells)
Age-only control [nats]	0.05 [-0.04, 0.15]	-0.04 [-0.09, 0.03]
Ridge HI lower bound [nats]	0.60 [0.47, 0.76]	0.31 [0.23, 0.43]
Charging features [nats]	0.91	0.48
Monotonicity: $\gamma = 0$, 0.995 (predicted)	0.65, 0.73 (0.70)	0.63, 0.72 (0.68)
Lower bound after shrinkage, $\gamma = 0.995$ [nats]	-0.00	-0.14
Worst-stage RMSE ratio, static / free	1.068	1.094
Protocol-blind / normalised estimator RMSE	0.190 / 0.167	0.188 / 0.179

80% of the nominal capacity are kept with the last recorded cycle as end of life if at least 100 cycles were recorded.

Number of units. For each size k (5, 10, 20, 40, 80 and 120, as far as k does not exceed 80% of the units), 10 random subsets of k units were drawn without replacement; on each subset the unit-grouped cross-validation (five folds; leave-one-unit-out for $k = 5$) of Section 5 was repeated with the age-only predictor and the ridge HI, and the lower bounds were computed with the conditional entropy re-estimated from the training units of each fold. A subset is valid when the gain in the information lower bound exceeds 0.1 nat and the age-only information lower bound is at most 0.05 nat. The scripts are `fleet_size_curve.py` and `fleet_size_battery.py`.

Template shrinkage. The template is the mean ridge HI at each cycle over the training engines of the fold, and the noise standard deviation is $\sigma_2 = \Delta m/(\sqrt{2}\cdot 1.645)$ with Δm the mean template increment over the first half of the median lifetime. Mon_∞ is predicted as the mean of $2\Phi_{\mathcal{N}}(\Delta m_k/(\sqrt{2}\sigma_2)) - 1$ over the template increments Δm_k up to the median lifetime, where $\Phi_{\mathcal{N}}$ is the standard normal distribution function. Because the template slope varies along life, the predicted limit is lower than for a constant slope.

Stages. Level and velocity are the summaries of the out-of-fold ridge HI, standardised over all pairs; the target is the RUL. Stages are bins of the life fraction (0.2 to 1, edges $0.2, 0.35, 0.5, 0.65, 0.8$; oracle stages) or equal-count bins of the age (observable stages). Within a stage, the free direction is the linear least-squares fit on both features, level-only and velocity-only use one feature, and a static direction (grid of 5°) is a fixed combination of the two features with a stage-specific affine map; the static direction minimises the sum of the out-of-fold MSEs over stages. Figure 4(c) gives the worst-stage RMSE ratio to the direction chosen separately at each stage.

Official test engines. The information lower bound uses the observed cycles of each test engine as its age, the lifetime distribution of the training engines, and the predictions of the methods of Section 6 (ten-seed ensembles for the learned encoders); the age-only baseline is the control. It assumes that the test truncation is independent of the lifetime, which the control checks.

S3. Efficiency estimation on real data

A direct estimate of $I(T; H)$ with the Kraskov–Stögbauer–Grassberger estimator [8] on rank-transformed window summaries gave, for several HIs, values above the estimate

for the 12-dimensional raw sensor summary, which is impossible for a true mutual information because the summary contains the HI. This reflects the bias of the estimator in 12 dimensions at a few hundred samples, a known limitation [9, 10]. A simulation with known mutual information (twelve Gaussian sensor features, a scalar target and a scalar HI with $\eta_{\text{info}} = 0.7$) shows that the estimator underestimates $I(T; \mathbf{X})$ strongly, so that $\hat{\eta}_{\text{info}}$ exceeds 1 in every replication even with 3000 samples. We therefore do not report η_{info} on real data. The information lower bound of Theorem 1(b) avoids high-dimensional estimation.

S4. Estimators used in the benchmark

This appendix describes the estimators compared in Section 6. The path-space objective follows from the theory; the architectures and the calibration are design choices.

S4.1. Path-space objective

We seek a causal stochastic encoder, the *physics-guided information bottleneck* (PGIB), $q_\theta(H_{1:t} \mid \mathbf{X}_{1:t}) = \prod_{k=1}^{t} q_\theta(H_k \mid H_{k-1}, \mathbf{X}_{1:k})$, whose increments are Gaussian, $\mathcal{N}\big(H_{k-1} + u_\theta(\mathbf{X}_{1:k})\Delta t,\ s_\theta^2(\mathbf{X}_{1:k})\Delta t\big)$, with a positive velocity u_θ (enforced by a softplus output) and a variance s_θ^2. The encoder is regularised towards a physics prior $p_\phi(H_{1:t}) = \prod_k \mathcal{N}\big(H_{k-1} + \nu(H_{k-1}; \phi)\Delta t,\ \sigma_B^2 \Delta t\big)$, $H_0 \equiv 0$, where Δt is the time step, ν a state-dependent drift with parameters ϕ, and σ_B the diffusion coefficient.

Proposition 6 (Compression bound and path-space KL)**.** *For any prior p_ϕ, $I(\mathbf{X}_{1:t}; H_{1:t}) \le \mathbb{E}_{\mathbf{X}}\big[D_{\text{KL}}(q_\theta(H_{1:t} \mid \mathbf{X}_{1:t}) \,\|\, p_\phi(H_{1:t}))\big]$, and for the Gaussian factorisation above the divergence is exactly*

$$D_{\text{KL}} = \frac{1}{2} \sum_{k=1}^{t} \mathbb{E}_{q_\theta(H_{k-1} \mid \mathbf{X}_{1:k-1})} \left[\frac{s_\theta^2}{\sigma_B^2} - 1 - \ln \frac{s_\theta^2}{\sigma_B^2} + \frac{(u_\theta - \nu(H_{k-1}; \phi))^2 \, \Delta t}{\sigma_B^2} \right]. \tag{5}$$

Proof. The first claim is the standard variational upper bound of the information bottleneck [11–13]: $I(\mathbf{X}; H) = \mathbb{E}_{\mathbf{X}} D_{\text{KL}}(q(H \mid \mathbf{X}) \| q(H)) \le \mathbb{E}_{\mathbf{X}} D_{\text{KL}}(q(H \mid \mathbf{X}) \| p(H))$ because the difference is $D_{\text{KL}}(q(H) \| p(H)) \ge 0$. The chain rule of the KL divergence [5, Sect. 2.5] over the causal factorisation gives $\sum_k \mathbb{E}_{q(H_{k-1} \mid \cdot)} D_{\text{KL}}(q(H_k \mid H_{k-1}, \mathbf{X}_{1:k}) \| p_\phi(H_k \mid H_{k-1}))$, and the Gaussian KL with means differing by $(u_\theta - \nu)\Delta t$ and variances $s_\theta^2 \Delta t$, $\sigma_B^2 \Delta t$ yields the summand (the Gaussian KL of the variational autoencoder [14]). □

The training objective is $\mathcal{L}_{\text{PGIB}} = \mathcal{L}_{\text{FHT}} + \beta \, D_{\text{KL}}$, where β weights the compression term and $\mathcal{L}_{\text{FHT}}$ is the negative log-likelihood of the observed failure time under the predictive distribution below, a variational surrogate for $-I(T_t; H_{1:t})$ in the sense of [12] (Algorithm 1 in Appendix S5).

S4.2. Predictive distribution

Let the posterior of the unit's drift μ given the HI increments be $\mathcal{N}(\hat{\mu}_t, \sigma_{\mu,t}^2)$ (conjugate update). Marginalising the inverse-Gaussian distribution over μ (a Gaussian integral; random-drift Wiener models are treated in [15, 16]) gives

$$p(T_t = \tau \mid H_{1:t}) = \frac{a}{\sqrt{2\pi\tau^3(\sigma_B^2 + \sigma_{\mu,t}^2 \tau)}} \exp\left(-\frac{(a - \hat{\mu}_t \tau)^2}{2\tau(\sigma_B^2 + \sigma_{\mu,t}^2 \tau)} \right), \qquad a = D_{\text{fail}} - \hat{D}_t. \tag{6}$$

Here σ_B^2 is the aleatoric and $\sigma_{\mu,t}^2\tau$ the epistemic contribution to the effective diffusion; as t grows $\sigma_{\mu,t}^2 \to 0$ and (6) reduces to the inverse-Gaussian distribution. It is a defective density, the missing mass being the probability that the threshold is never reached, which can happen only for a non-positive drift, and $\hat{D}_t$ is the filtered state rather than the raw HI value, which avoids the degeneracies of non-monotone paths. The distribution was checked against numerical integration of the mixture (Section 3).

S4.3. Observation-corrected and regime-aware encoders

The encoder above integrates a positive velocity, $H_k = H_{k-1} + u_\theta \Delta t + \ldots$, so nothing anchors the level to the observations (Section 4.4). The observation-corrected encoder, PGIB-KF (KF: Kalman filter), therefore adds a direct level estimate $l_k = f_l(\mathbf{h}_k)$ and a learned gain $\omega_k = \mathrm{sig}(f_\omega(\mathbf{h}_k)) \in (0,1)$ to the encoder features $\mathbf{h}_k$ (here sig is the logistic function and $\epsilon_k \sim \mathcal{N}(0,1)$ the encoder noise),

$$H_k = (1-\omega_k)\big(H_{k-1} + u_k + s_k\epsilon_k\big) + \omega_k\, l_k. \tag{7}$$

Given H_{k-1} this is again a Gaussian transition with drift $u_k^{\mathrm{eff}} = (1-\omega_k)u_k + \omega_k(l_k - H_{k-1})$ and standard deviation $s_k^{\mathrm{eff}} = (1-\omega_k)s_k$, so Proposition 6 applies unchanged: the bottleneck now prices the use of the observation beyond what the physics prior allows ($\beta \to \infty$ gives $\omega \to 0$, a pure integrator). The HI is no longer monotone by construction.

The error accumulation of an integrating encoder and its correction by the gain ω are analysed in Proposition 2.

By Proposition 1, a regime-blind estimator has an error floor, and per-regime normalisation removes the offsets but not the gains. The regime-aware variant, *PGIB-KFR*, therefore places the regime in the *observation* model, where the proposition locates it. The process model is shared, whereas the level estimate l_k and the gain ω_k use regime-specific linear heads on the encoder features, $l_k = \mathbf{h}_k^\top \mathbf{W}_{r_k}^l + b_{r_k}^l$ and $\omega_k = \mathrm{sig}(\mathbf{h}_k^\top \mathbf{W}_{r_k}^\omega + b_{r_k}^\omega)$, where r_k is the regime, which is also an input of the encoder. The regimes are the clusters of the standardised operating settings. With a single regime PGIB-KFR reduces to PGIB-KF, which we check on the single-regime subsets.

S4.4. Calibrated predictive distribution with age fusion

Because the age is always known (Theorem 1(c)), the HI-based distribution $p_{\mathrm{HI}}(\tau)$, which is (6) with diffusion $c\,v_t$ and drift variance $c\,v_t/w_t$, is pooled log-linearly [17, 18] with an age-based lifetime distribution. Here v_t is the variance of the HI increments over the trailing window of length w_t for non-learned HIs and the learned σ_B^2 for the bottleneck models, and $c > 0$ is a calibration multiplier. The age-based distribution is $p_{\mathrm{age}}(\tau \mid t) = f_L(t+\tau)/S_L(t)$, the log-normal lifetime density f_L with survival function S_L [19], fitted on training lifetimes (log-scale floor 0.15) conditioned on survival to age t. The pooled distribution is

$$p(\tau) \;\propto\; p_{\mathrm{HI}}(\tau)^{w_{\mathrm{h}}}\, p_{\mathrm{age}}(\tau \mid t)^{w_{\mathrm{a}}}, \qquad (c, w_{\mathrm{h}}, w_{\mathrm{a}}) \in \mathcal{G}, \tag{8}$$

on a finite grid $\mathcal{G}$ ($c \in [10^{-2}, 10^2]$ on a log scale, ten pairs $(w_{\mathrm{h}}, w_{\mathrm{a}})$ including $(1,0)$ and $(0,1)$). The parameters minimise the negative log-likelihood of the true RUL, a strictly proper scoring rule [20], on HI paths produced *out of sample*: each HI construction is fitted on all but one training unit and applied to the left-out unit; the test unit is never used. The family contains the plain HI-based distribution and the age-only distribution, so in-sample the pooled distribution cannot do worse than either. Weights below one temper

the corresponding distribution and widen the predictive interval, which the otherwise under-covering plain distribution requires. The pooling is a calibration procedure, not a Bayes-exact fusion, which would also require conditional independence of the HI and the age given the RUL, and division by the marginal distribution of the RUL. The same predictor is used for every HI, and a variant without fusion ($w_a = 0$) is reported for each HI, so that the effects of the HI and of the fusion can be separated.

S5. Data, protocols and evidence tiers

Data. Public run-to-failure collections were used: NASA PCoE cells B0005, B0006, B0007 and B0018 [21]; CALCE CS2 cells [22]; the FEMTO-ST/PRONOSTIA bearings [23]; the NASA IMS bearings [24, 25]; and the C-MAPSS turbofan engines [26]. For batteries, the capacity is used only to define the EOL and is never an input. The EOL is the first cycle at which the rolling-median capacity falls to 75% (NASA) or 80% (CALCE) of the initial capacity; an absolute threshold of 1.4 Ah was not used because cell B0007 never reaches it. The lifetimes are 127, 73, 162 and 100 cycles (NASA) and 542, 499, 572 and 600 cycles (CS2_35 to CS2_38); the data are truncated at the EOL, and CALCE features are averaged over blocks of 4 cycles. Six per-cycle features are extracted from the charge and discharge records: constant-current time, constant-voltage time, height and voltage of the incremental-capacity peak, mean discharge temperature (NASA) or voltage drop at discharge onset (CALCE), and the discharge voltage at a fixed time (600 s for NASA, 1200 s for CALCE).

Further cells were screened by inclusion rules fixed before running any model (scripts and outputs are supplied). A NASA cell had to be at 24°C, have at least 100 discharge cycles, be discharged at a roughly constant 1.4 to 2.1 A, start between 1.5 and 2.1 Ah and reach the 75% EOL; only the four cells above qualify. Among the CALCE cells, CS2_5, 24 and 25 are partial-discharge tests, CS2_9 is discharged at a rate of 2C (twice the nominal capacity per hour), and CS2_3, 6 and 7 are short or follow other protocols, so none was used. CS2_33 and CS2_34 have the same chemistry and run to the EOL (485 and 479 cycles) but are discharged at 0.55 A; they are used only in a separate six-cell experiment, labelled protocol-heterogeneous, and the four-cell set remains the primary one.

For bearings, each run ends at failure, so EOL is the last snapshot. Condition 1 (nominally 1800 rpm, 4000 N; Bearing1_1–1_7) and condition 2 (1650 rpm, 4200 N; Bearing2_1–2_7) are evaluated separately, each with 7 units from the learning and the full test sets; condition 3 has only 3 units and was not used. The lifetimes are 2803, 871, 2375, 1428, 2463, 2448, 2259 snapshots (condition 1) and 911, 797, 1955, 751, 2311, 701, 230 snapshots (condition 2), each snapshot covering 0.1 s of 25.6 kHz acceleration every 10 s. Eight features are computed per snapshot: logarithm of the root mean square (RMS), kurtosis, crest factor and spectral entropy of the horizontal and of the vertical channel. Features are averaged over blocks of 20 (condition 1) or 8 (condition 2) snapshots, and the RUL is reported in snapshots.

Evaluation is leave-one-unit-out: each unit in turn is held out, the remaining units are used for training (three for NASA and four-cell CALCE, five for six-cell CALCE, six for each bearing condition), and metrics are computed at every step in the window 40–90% of life and pooled over the held-out units.

Further collections. FEMTO condition 3 (1500 rpm, 5000 N; Bearing3_1–3_3, with 515, 1637 and 434 snapshots, block length 10) and the NASA IMS bearings were reserved and

used only as confirmation sets. For IMS the four bearings that failed (Test 1 bearings 3 and 4, Test 2 bearing 1, Test 3 bearing 3; 2156, 2156, 984 and 6324 snapshots of 1 s at 20 kHz every 10 min) are used with one accelerometer channel each and four features per snapshot (log RMS, kurtosis, crest factor, spectral entropy), averaged over blocks of 20 snapshots. For C-MAPSS we use the single-condition subsets FD001 and FD003 (100 training engines run to failure and 100 truncated test engines with the official RUL labels each); the 14 informative sensors are standardised with training statistics. The 100 training engines are split by identifier into 80 engines to fit the HI constructions and 20 (identifier divisible by five) to select the bottleneck configuration and to calibrate the predictor; the test engines are predicted once at their last observed cycle, and 95% bootstrap intervals are computed over engines. The multi-regime subsets FD002 (260 training and 259 test engines) and FD004 (249 and 248) have six operating regimes; for them the only change is that sensors are standardised within regime, the six regimes being the clusters of the standardised operating settings found on the fitting engines (the clusters recover the six discrete settings exactly), the same preprocessing being applied to every method; the split is by identifier as above, with 208 and 200 fitting engines.

Methods. To isolate the effect of HI construction, every HI is mapped to the same scale and passed to the same calibrated predictor of Appendix S4.4. The sensor features are standardised, and each HI is mapped by an affine map $H \mapsto (H - H_{\mathrm{s}})/(H_{\mathrm{e}} - H_{\mathrm{s}})$, where H_{s} and H_{e} are the averages, over the *training* units only, of the mean of the first five and of the last five HI values of a unit (the end of life of a training unit is known); the standardisation, the PCA direction, the combination weights of the Geo-opt HI, the encoders and the affine map are all fitted on the training units and applied unchanged to the test unit, whose own end-of-life value is never used. The PGIB scale is set by its end-of-life penalty on the training units. The predictor of Appendix S4.4 then uses: the mixed inverse-Gaussian distribution (6) with the drift taken from the slope of the smoothed HI over a trailing window of $\max(10, 0.25t)$ steps, pooled with the age-based distribution. The parameters $(c, w_{\mathrm{h}}, w_{\mathrm{a}})$ are chosen on HI paths that are out of sample for the HI construction (leave-one-training-unit-out for the small sets, the 20 calibration engines for C-MAPSS). Implementation details of the compared HIs: the Geo-opt HI is found by differential evolution; the β-VAE uses a causal gated recurrent unit (GRU) encoder with a one-dimensional latent variable and a Gaussian prior; VIB is the PGIB with a zero-drift prior, that is, without physics; the PGIB has the state-dependent drift $\nu(H; \phi) = \phi_1 \exp(\phi_2 H)$ (Appendix S4.3 for PGIB-KF). For the bottleneck models, the compression weight $\beta \in \{0.03, 0.3, 3, 30\}$ and the encoder (GRU or causal feed-forward window network) are selected by inner leave-one-training-unit-out validation of the predictive negative log-likelihood, with at most three inner folds when more than five training units are available; for PGIB-KF only $\beta \in \{0.03, 0.3, 3\}$ is searched. The held-out unit is never used. Learned models use three seeds, and the other HIs are deterministic. The sensor-free reference is the age-based distribution p_{age} of Appendix S4.4. The metrics are the RMSE and the mean absolute error (MAE) of the median prediction, the prediction interval coverage probability (PICP) and normalised average width (PINAW) of the 95% interval, and the continuous ranked probability score (CRPS) [20]. For C-MAPSS the asymmetric score is $\sum_i (e^{-d_i/13} - 1)$ for $d_i < 0$ and $\sum_i (e^{d_i/10} - 1)$ for $d_i \geq 0$, summed over engines, with d_i the predicted minus the true RUL [26].

Table S2: Algorithm 1: training of the PGIB and Bayesian FHT prognostics (Appendix S4).

Input: training sequences $\{\mathbf{X}_{1:T_n}^{(n)}, T^{(n)}\}$, threshold $D_{\text{fail}} = 1$, compression weight β, step Δt. **Output:** q_{θ^*}, (ϕ^*, σ_B^*), predictive distribution (6).
1. Causal encoder (temporal convolutional network or GRU): $u_\theta, s_\theta \leftarrow \text{softplus}(\cdot)$ from $\mathbf{X}_{1:k}$.
2. Sample the path by reparametrisation: $H_k = H_{k-1} + u_\theta \Delta t + s_\theta \sqrt{\Delta t}\, \epsilon_k$, $\epsilon_k \sim \mathcal{N}(0,1)$, $H_0 = 0$.
3. Conjugate drift update from $\Delta H_{1:t}$ to obtain $(\hat{\mu}_t, \sigma_{\mu,t}^2)$.
4. $\mathcal{L}_{\text{FHT}} = -\ln p(T_t \mid H_{1:t})$ from (6); $\mathcal{L}_{\text{KL}}$ from Proposition 6.
5. Update (θ, ϕ, σ_B) with the AdamW optimiser on $\mathcal{L}_{\text{FHT}} + \beta \mathcal{L}_{\text{KL}}$ (plus a penalty on $H_{T_{\text{EOL}}} - D_{\text{fail}}$ at the EOL).

Protocol history and evidence tiers. The study proceeded in five protocols, which determine how the numbers can be read. P1 (Appendix S7) showed that the integrating encoder under-covers and that the age-only baseline is competitive. P2 added out-of-sample calibration, age fusion and a prediction grid independent of the test unit; it was frozen, with the file hash recorded, before it was run on the confirmation sets (FEMTO condition 3 and IMS). On C-MAPSS FD001, P2 showed that the integrating encoder is considerably worse than simple HIs even with 80 training engines. The diagnosis of Proposition 2 led to P3, the observation-corrected PGIB-KF, which was frozen and run once on FD003, a subset not used before, and then, with regime-wise preprocessing, once on FD002 and FD004. (A first launch used only the first 100 training engines because the identifier list was fixed in the script; it was stopped before any result was produced, and the script and the recorded hash were corrected.) P4 added the regime-aware encoder and a regime-specific ridge baseline and was run once on each subset, the choice between PGIB-KF and PGIB-KFR and of β being made on the calibration engines only; FD002 and FD004 are therefore development sets for P4, and no independent multi-regime data set was available (the successor N-CMAPSS has only six to ten units per subset). Because the differences between learned encoders and simple HIs were of the size of the seed-to-seed variability, P5 retrained the learned encoders on the four subsets with ten fresh seeds, keeping architecture, β and calibration fixed; it supersedes the three-seed results where they disagree. The evidence tiers are: (i) *independent confirmation*: C-MAPSS FD003, FD002 and FD004 for P3 (not for P4), and FEMTO condition 3 and IMS for P2 (not for P3); (ii) *development*: NASA, CALCE (four and six cells), FEMTO conditions 1 and 2 and C-MAPSS FD001, which informed the design and whose results are therefore not unbiased estimates of out-of-sample performance.

S6. Detailed benchmark results

This appendix gives the tables behind Section 6: protocol P2 on the development and confirmation sets (Tables S3 and S4), the C-MAPSS subsets (Tables S5, S6, S7 and S8), the effect of the age fusion (Table S9), an ablation of the integrating PGIB (Table S10) and the information lower bounds (Tables S11 and S12). Protocols and evidence tiers are described in Appendix S5.

Table S3: Protocol P2, development sets. RMSE, MAE and CRPS in cycles (batteries) or 10 s snapshots (bearings); mean ± standard deviation over seeds; bold: lowest RMSE.

NASA PCoE (4 cells)					
Method	RMSE	MAE	PICP [%]	PINAW	CRPS
Age-only (no sensors)	34.4	28.7	84	2.68	21.0
PCA-HI	24.9	19.3	95	2.46	13.8
Geo-opt HI	**24.3**	20.1	93	3.22	17.6
β-VAE-HI	31.7±2.2	25.3±1.8	98±0	4.24±0.17	20.0±0.9
VIB-HI (no physics)	33.9±1.2	28.3±1.2	90±1	5.40±0.03	27.1±0.7
PGIB-HI (ens.)	34.8	29.0	93	6.51	27.3
PGIB-KF-HI (ens.)	44.3	39.4	84	4.55	29.1

CALCE CS2 (4 cells)					
Method	RMSE	MAE	PICP [%]	PINAW	CRPS
Age-only (no sensors)	46.5	40.0	100	1.13	28.6
PCA-HI	51.3	44.0	100	1.06	30.2
Geo-opt HI	43.3	35.7	98	0.98	26.5
β-VAE-HI	49.9±0.9	41.6±1.0	100	1.09±0.00	29.5±0.4
VIB-HI (no physics)	48.3±1.4	40.6±0.9	83±4	0.85±0.00	30.8±0.8
PGIB-HI (ens.)	**38.4**	34.2	99	0.74	23.3
PGIB-KF-HI (ens.)	50.3	36.6	77	0.83	33.0

CALCE CS2 (6 cells)					
Method	RMSE	MAE	PICP [%]	PINAW	CRPS
Age-only (no sensors)	50.6	45.5	100	1.13	30.1
PCA-HI	56.8	50.0	100	1.09	32.8
Geo-opt HI	**46.3**	43.2	100	1.02	27.6
β-VAE-HI	73.5±4.9	60.9±3.9	93±3	1.04±0.02	42.4±3.2
VIB-HI (no physics)	54.7±1.7	48.7±1.5	92±1	0.91±0.01	32.4±1.2
PGIB-HI (ens.)	48.9	44.8	90	0.87	30.8
PGIB-KF-HI (ens.)	71.9	63.6	84	0.92	45.2

FEMTO cond. 1 (7 bearings)					
Method	RMSE	MAE	PICP [%]	PINAW	CRPS
Age-only (no sensors)	**495.8**	371.4	94	3.45	303.7
PCA-HI	786.9	621.6	72	4.72	489.2
Geo-opt HI	533.2	416.8	96	3.19	311.5
β-VAE-HI	737.1±191.7	525.4±102.0	85±6	4.84±0.04	421.4±87.9
VIB-HI (no physics)	584.1±29.4	467.0±26.4	82±5	3.48±0.13	357.9±29.0
PGIB-HI (ens.)	534.7	401.8	84	4.67	340.9

FEMTO cond. 2 (7 bearings)					
Method	RMSE	MAE	PICP [%]	PINAW	CRPS
Age-only (no sensors)	**430.6**	373.5	97	10.38	278.6
PCA-HI	558.3	450.5	67	8.52	391.1
Geo-opt HI	504.9	414.7	85	11.22	396.1
β-VAE-HI	531.2±11.3	434.6±20.1	88±2	9.78±0.02	368.4±2.0
VIB-HI (no physics)	448.2±0.4	337.7±1.3	97	11.39±0.14	318.5±2.6
PGIB-HI (ens.)	449.5	333.6	97	13.97	341.2

Table S4: Protocol P2, confirmation sets (snapshots). PGIB-KF is exploratory here.

FEMTO cond. 3 (3 bearings)					
Method	RMSE	MAE	PICP [%]	PINAW	CRPS
Age-only (no sensors)	563.6	529.7	37	10.60	460.2
PCA-HI	507.5	418.4	32	6.50	398.8
Geo-opt HI	567.1	525.1	30	12.82	453.4
β-VAE-HI	**462.0±20.8**	389.5±23.6	37	8.86±0.72	363.4±20.9
VIB-HI (no physics)	505.0±0.1	438.3±0.3	37	12.55±0.00	442.5±0.2
PGIB-HI (ens.)	510.3	450.0	37	12.56	453.5
PGIB-KF-HI (ens.)	555.0	479.3	33	9.10	438.6

NASA IMS (4 bearings)					
Method	RMSE	MAE	PICP [%]	PINAW	CRPS
Age-only (no sensors)	1692.8	1503.1	83	10.78	1106.3
PCA-HI	**745.2**	605.7	96	8.05	498.0
Geo-opt HI	942.7	702.8	99	16.31	771.6
β-VAE-HI	2769.6±282.2	1324.8±242.7	84±6	10.31±0.71	1184.6±197.8
VIB-HI (no physics)	1884.6±10.5	1434.4±12.4	91	16.41±0.01	1122.5±8.8
PGIB-HI (ens.)	1623.8	1288.1	98	16.49	1017.5
PGIB-KF-HI (ens.)	1852.3	1519.8	43	12.29	1545.6

Table S5: C-MAPSS FD003 (independent confirmation set of P3; PGIB-KFR and regime-specific ridge rows are development results). RMSE: single models, mean over seeds; seed-avg.: RMSE of the seed-averaged prediction with 95% bootstrap interval over engines.

Method	RMSE	RMSE seed-avg. [95% CI]	MAE	Score	PICP [%]	PINAW	CRPS
Age-only (no sensors)	47.6	42.3–52.6	39.6	114536	99	1.91	28.6
PCA-HI	33.8	28.4–38.3	25.5	10747	97	1.47	19.7
Geo-opt HI	21.9	17.8–26.0	15.9	1864	100	1.19	13.1
Ridge-HI (supervised)	26.8	22.9–30.6	21.0	6147	93	1.20	16.2
Ridge-HI, regime-specific	26.8	22.8–30.6	21.0	6147	93	1.20	16.2
β-VAE-HI	40.6±1.1	34.9–45.6	32.0±0.7	29236	99	1.69	23.8±0.3
VIB-HI (no physics prior)	37.9±0.8	31.7–42.2	29.3±0.8	18680	96	2.16	24.2±0.6
PGIB-HI (GRU/window encoder)	39.9±1.0	33.9–44.4	31.2±0.4	21945	95	1.98	24.3±0.3
PGIB-HI, 3-seed ensemble	39.5	34.1–45.0	31.1	17643	95	1.98	24.1
PGIB-KF-HI	24.2±2.5	16.4–23.1	18.6±1.5	2296	94	1.19	14.8±0.4
PGIB-KF-HI, 3-seed ensemble	19.4	16.0–22.8	14.6	790	97	1.20	12.2
PGIB-KFR-HI (regime-aware)	27.4±0.1	21.8–29.3	21.2±0.3	5417	95	1.24	15.6±0.4
PGIB-KFR-HI, 3-seed ensemble	25.4	21.6–29.1	20.3	4170	97	1.24	14.6

Table S6: C-MAPSS FD001 (development set; same layout as Table S5).

Method	RMSE	RMSE seed-avg. [95% CI]	MAE	Score	PICP [%]	PINAW	CRPS
Age-only (no sensors)	36.6	32.4–40.8	30.5	8023	99	1.18	21.1
PCA-HI	21.8	19.1–24.4	17.9	908	96	0.83	12.6
Geo-opt HI	22.0	19.5–24.3	18.0	953	96	0.83	12.7
Ridge-HI (supervised)	26.0	22.0–29.8	19.7	1606	97	0.85	14.1
Ridge-HI, regime-specific	26.0	22.0–29.8	19.7	1606	97	0.85	14.1
β-VAE-HI	22.2±0.4	19.0–24.5	18.0±0.5	875	95	0.82	12.7±0.3
VIB-HI (no physics prior)	34.2±0.3	28.8–38.3	27.1±0.7	7386	91	0.96	19.3±0.4
PGIB-HI (GRU/window encoder)	35.6±1.0	29.8–40.9	26.9±0.6	11524	94	0.90	19.6±0.4
PGIB-HI, 3-seed ensemble	35.4	29.9–40.9	26.6	10874	94	0.90	19.4
PGIB-KF-HI	27.0±2.2	18.8–27.1	20.2±1.2	2868	82	0.56	14.9±1.1
PGIB-KF-HI, 3-seed ensemble	22.9	18.6–26.9	16.6	1309	93	0.56	12.1
PGIB-KFR-HI (regime-aware)	26.9±3.6	21.8–31.0	20.2±2.6	2937	94	0.64	14.4±2.1
PGIB-KFR-HI, 3-seed ensemble	26.5	22.0–30.9	19.5	2094	96	0.64	13.8

Table S7: C-MAPSS FD002 (multi-regime; confirmation set for P3, development set for P4; same layout as Table S5).

Method	RMSE	RMSE seed-avg. [95% CI]	MAE	Score	PICP [%]	PINAW	CRPS
Age-only (no sensors)	38.3	35.2–41.2	31.4	53131	97	0.82	21.6
PCA-HI	28.7	26.0–31.4	21.9	10331	95	0.66	15.7
Geo-opt HI	30.5	27.0–33.9	21.8	16164	98	0.59	15.2
Ridge-HI (supervised)	29.4	26.1–32.6	21.7	11935	94	0.58	15.1
Ridge-HI, regime-specific	28.4	25.3–31.5	20.9	11149	94	0.58	14.7
β-VAE-HI	30.6±0.8	27.3–33.1	24.0±0.7	11996	92	0.65	17.0±0.4
VIB-HI (no physics prior)	36.4±0.4	33.3–39.4	29.8±0.2	26118	82	0.68	21.1±0.1
PGIB-HI (GRU/window encoder)	36.8±0.2	33.6–39.4	30.1±0.5	28553	83	0.68	21.4±0.5
PGIB-HI, 3-seed ensemble	36.6	33.7–39.4	29.9	26625	82	0.68	21.3
PGIB-KF-HI	31.3±1.5	27.8–33.1	25.4±1.5	14139	81	0.79	18.6±1.1
PGIB-KF-HI, 3-seed ensemble	30.4	27.8–32.9	24.5	10983	81	0.79	18.0
PGIB-KFR-HI (regime-aware)	30.7±2.0	27.2–32.2	24.8±2.6	14802	86	0.91	18.5±1.9
PGIB-KFR-HI, 3-seed ensemble	29.6	27.0–32.1	23.7	13089	87	0.91	17.8

Table S8: C-MAPSS FD004 (multi-regime; confirmation set for P3, development set for P4; same layout as Table S5).

Method	RMSE	RMSE seed-avg. [95% CI]	MAE	Score	PICP [%]	PINAW	CRPS
Age-only (no sensors)	54.7	50.9–58.5	46.0	611896	95	1.31	31.8
PCA-HI	44.3	40.7–48.0	35.4	104864	92	1.03	24.9
Geo-opt HI	29.7	26.6–32.7	21.9	9468	100	0.86	16.1
Ridge-HI (supervised)	30.9	28.1–33.4	23.8	11632	94	1.14	18.5
Ridge-HI, regime-specific	31.1	28.2–33.9	24.0	10794	92	1.14	18.6
β-VAE-HI	48.1±3.1	42.7–50.2	39.0±3.0	294449	93	1.17	27.7±1.9
VIB-HI (no physics prior)	51.8±1.0	47.3–56.2	41.9±0.6	697542	90	1.13	29.5±0.4
PGIB-HI (GRU/window encoder)	49.9±2.4	44.2–53.9	39.5±1.5	1143425	90	1.42	29.3±0.7
PGIB-HI, 3-seed ensemble	49.5	44.8–54.3	38.9	1085286	91	1.42	28.9
PGIB-KF-HI	42.1±2.8	37.2–43.6	33.9±1.9	241870	78	1.08	25.6±1.3
PGIB-KF-HI, 3-seed ensemble	40.6	37.3–44.0	32.9	124580	79	1.07	24.7
PGIB-KFR-HI (regime-aware)	37.1±2.5	31.4–37.2	29.4±1.5	187056	85	1.12	22.3±0.8
PGIB-KFR-HI, 3-seed ensemble	34.8	31.8–37.8	27.8	37063	85	1.12	21.0

Table S9: Effect of the age fusion: RMSE with fusion / without fusion (after recalibration with $w_a = 0$).

HI	NASA	CALCE-4	CALCE-6	FEMTO-1	FEMTO-2	FEMTO-3	IMS
PCA-HI	25 / 18	51 / 186	57 / 184	787 / 1470	558 / 564	507 / 503	745 / 745
Geo-opt HI	24 / 32	43 / 131	46 / 126	533 / 1083	505 / 489	567 / 437	943 / 943
β-VAE-HI	32 / 26	50 / 110	73 / 270	737 / 1421	531 / 457	462 / 396	2770 / 2770
VIB-HI (no physics)	34 / 30	48 / 59	55 / 106	584 / 681	448 / 469	505 / 507	1885 / 1885
PGIB-HI (ens.)	35 / 30	38 / 35	49 / 85	535 / 754	449 / 445	510 / 509	1624 / 1624
PGIB-KF-HI (ens.)	44 / 44	50 / 59	72 / 88	–	–	555 / 556	1852 / 1852

Table S10: Ablation of the integrating PGIB on C-MAPSS FD003 (age-fused predictor; mean ± standard deviation over three seeds). PICP, PINAW: coverage and normalised width of the 95% interval; CRPS: continuous ranked probability score.

Variant	RMSE	MAE	PICP [%]	PINAW	CRPS
Full PGIB	39.9±1.0	31.2±0.4	95±1	1.98±0.03	24.3±0.3
Without bottleneck ($\beta = 0$)	39.1±1.9	30.7±1.0	96±0	1.99±0.04	23.9±0.7
Without physics prior (VIB)	37.9±0.8	29.3±0.8	96±0	2.16±0.05	24.2±0.6
Without epistemic drift variance	45.9±1.4	37.3±1.4	92±1	1.32±0.09	26.1±0.8

Table S11: Protocol P2, lower bound on the information implied by the achieved error (nats); < 0: uninformative. $h(T)$: entropy of the pooled held-out RUL.

Dataset	$h(T)$	Age-only	PCA	Geo-opt.	β-VAE	VIB	PGIB	PGIB-KF
NASA	4.36	< 0	< 0	< 0	< 0	< 0	< 0	< 0
CALCE-4	5.67	0.41	0.31	0.48	0.34	0.37	0.60	0.33
CALCE-6	5.66	0.31	0.20	0.40	< 0	0.24	0.35	< 0
FEMTO-1	7.24	< 0	< 0	< 0	< 0	< 0	< 0	–
FEMTO-2	7.04	< 0	< 0	< 0	< 0	< 0	< 0	–
FEMTO-3	6.70	< 0	< 0	< 0	< 0	< 0	< 0	< 0
IMS	8.05	< 0	0.02	< 0	< 0	< 0	< 0	< 0

Table S12: Age-conditional information lower bound (nats) under a fitted log-normal lifetime model; $\bar{h}(T \mid t)$: mean conditional entropy. A positive value for the age-only predictor shows that the model overestimates $h(T \mid t)$.

Data set	$\bar{h}(T \mid t)$	Age-only	PCA	Geo-opt.	β-VAE	VIB	PGIB
NASA	4.67	-0.28	0.04	0.07	-0.14	-0.23	-0.29
CALCE-4	5.74	0.49	0.39	0.56	0.42	0.46	0.68
CALCE-6	5.69	0.35	0.23	0.44	-0.05	0.31	0.38
FEMTO-1	7.81	0.19	-0.28	0.11	-0.15	-0.03	0.11
FEMTO-2	7.77	0.29	0.03	0.13	0.08	0.25	0.24
FEMTO-3	7.30	-0.45	-0.35	-0.46	-0.19	-0.34	-0.35
IMS	8.65	-0.20	0.62	0.39	-0.58	-0.31	-0.16

S7. Initial benchmark protocol (P1)

The first benchmark used a simpler protocol (P1). Every HI was calibrated on in-sample training paths, the plain first-hitting-time distribution (6) was used without age fusion, and the predictive grid extended to three times the held-out unit's own life, which leaks the length of the test unit into the support of the predictive distribution (removed in P2). The pilot on NASA B0005, the results below and the observation that the age-only baseline is competitive motivated P2. The P1 results are reported for completeness and are not used for the conclusions of the main text.

Pilot run. A pilot on the NASA B0005 fold, with a GRU-only encoder and $\beta \in \{0.003, 0.03, 0.3\}$, gave a PGIB RMSE of 56.3 cycles and zero interval coverage, with the selected β at the edge of the grid. The selection grid was therefore extended and the feed-forward encoder added as a candidate. Both changes are chosen per fold by inner validation, but the decision to introduce them was made after seeing a held-out result for one fold. No baseline was modified, and no change was made after the full runs reported below; the bearing experiments were run once with the final protocol.

Results under P1. The outcome depends on the data set (Tables S13 and S14, Figure S1), and the age-only baseline is needed to read it. On NASA the RMSE of the age-only baseline is 39.8 cycles; PCA-HI (17.8), the Geo-opt HI (21.9), the β-VAE (30.8) and VIB (33.3) outperform it, whereas the PGIB (47.6) does not, and its intervals cover only 42% of the true RULs. On CALCE the RMSE of the age-only baseline is 49.3; only the PGIB (33.0) outperforms it clearly, while PCA-HI, the Geo-opt HI and the β-VAE (117 to 164) are considerably worse than the age-only baseline. With the two 0.5C-discharge cells added, the PGIB keeps the lowest RMSE (50.4) but its margin over the age-only baseline

Table S13: Protocol P1, battery benchmarks, leave-one-unit-out (mean ± standard deviation over seeds). RMSE, MAE and CRPS in cycles.

NASA PCoE (leave-one-unit-out)								
Method	Mon	Tre	Pro	RMSE	MAE	PICP [%]	PINAW	CRPS
Age-only reference (no sensors)	–	–	–	39.8	34.7	–	–	–
PCA-HI	0.27	0.85	0.77	17.8	14.0	97	2.52	11.1
Geo-opt HI	0.21	0.93	0.95	21.9	17.0	90	2.43	12.8
β-VAE-HI	0.30±0.01	0.71±0.00	0.95±0.01	30.8±0.8	26.0±0.5	95±1	2.06±0.08	17.6±0.5
VIB-HI (no physics prior)	1.00	1.00±0.00	0.67±0.03	33.3±2.8	27.7±2.1	88±4	3.47±0.10	20.5±2.3
PGIB-HI	1.00	1.00±0.00	0.67±0.01	47.6±2.1	42.3±1.9	42±3	2.39±0.15	34.3±1.5

CALCE CS2 (4 cells, 1C/1C) (leave-one-unit-out)								
Method	Mon	Tre	Pro	RMSE	MAE	PICP [%]	PINAW	CRPS
Age-only reference (no sensors)	–	–	–	49.3	43.8	–	–	–
PCA-HI	0.05	0.72	0.83	130.5	110.4	92	2.74	82.3
Geo-opt HI	0.39	0.95	0.98	117.2	86.8	90	3.20	69.2
β-VAE-HI	0.02±0.01	0.72±0.01	0.80±0.01	163.7±0.7	148.0±0.9	83±1	1.87±0.02	111.1±0.7
VIB-HI (no physics prior)	1.00	1.00±0.00	0.92±0.01	53.2±0.9	47.2±1.4	71±4	1.24±0.13	38.4±0.7
PGIB-HI	1.00	1.00±0.00	0.95±0.01	33.0±4.9	29.9±3.9	55±8	0.34±0.02	22.6±3.1

CALCE CS2 (6 cells, incl. 0.5C-discharge CS2_33/34) (leave-one-unit-out)								
Method	Mon	Tre	Pro	RMSE	MAE	PICP [%]	PINAW	CRPS
Age-only reference (no sensors)	–	–	–	54.7	50.0	–	–	–
PCA-HI	0.13	0.28	0.21	191.8	172.0	50	0.99	149.6
Geo-opt HI	0.39	0.95	0.98	106.9	82.7	95	3.11	63.7
β-VAE-HI	0.05±0.02	0.52±0.10	0.33±0.04	201.6±5.3	179.9±5.0	48±0	1.46±0.18	156.1±3.1
VIB-HI (no physics prior)	1.00	1.00±0.00	0.87±0.02	69.7±6.0	54.4±8.8	90±5	2.24±0.23	44.9±4.8
PGIB-HI	1.00	1.00±0.00	0.91±0.01	50.4±4.4	44.7±5.1	69±5	0.58±0.02	33.4±4.2

Table S14: Protocol P1, FEMTO bearing benchmarks. RMSE, MAE and CRPS in 10 s snapshots.

FEMTO bearings, condition 1 (7 units) (leave-one-unit-out)								
Method	Mon	Tre	Pro	RMSE	MAE	PICP [%]	PINAW	CRPS
Age-only reference (no sensors)	–	–	–	602.5	515.4	–	–	–
PCA-HI	0.06	0.56	0.70	784.5	663.2	93	2.80	485.7
Geo-opt HI	0.04	0.29	0.39	770.2	623.3	91	3.34	459.1
β-VAE-HI	0.07±0.02	0.58±0.01	0.73±0.01	779.8±51.1	664.4±35.6	91±3	2.69±0.13	492.4±20.6
VIB-HI (no physics prior)	1.00	1.00±0.00	0.71±0.02	570.1±56.0	471.0±50.7	78±5	3.04±0.21	365.6±44.1
PGIB-HI	1.00	1.00±0.00	0.71±0.02	524.5±45.3	423.6±37.3	75±5	2.65±0.10	336.5±31.8

FEMTO bearings, condition 2 (7 units) (leave-one-unit-out)								
Method	Mon	Tre	Pro	RMSE	MAE	PICP [%]	PINAW	CRPS
Age-only reference (no sensors)	–	–	–	664.4	584.6	–	–	–
PCA-HI	0.04	0.43	0.17	613.9	487.6	54	2.25	433.9
Geo-opt HI	0.06	0.37	0.46	559.2	457.1	73	2.33	356.5
β-VAE-HI	0.06±0.01	0.46±0.05	0.42±0.16	539.8±31.7	439.6±24.6	70±2	2.01±0.06	348.6±26.1
VIB-HI (no physics prior)	1.00	0.99±0.00	0.39±0.02	616.8±1.5	493.8±1.5	41±1	2.62±0.23	451.1±3.2
PGIB-HI	1.00	0.99±0.00	0.46±0.02	621.8±4.3	498.2±8.1	36±2	2.25±0.01	464.3±7.9

Table S15: Protocol P1, lower bound on the information implied by the achieved error (Theorem 1(b)), in nats; < 0: uninformative. $h(T)$: entropy of the pooled held-out RUL.

Dataset	$h(T)$	PCA	Geo-opt.	β-VAE	VIB	PGIB
NASA	4.36	0.06	< 0	< 0	< 0	< 0
CALCE (4)	5.67	< 0	< 0	< 0	0.27	0.76
CALCE (6)	5.66	< 0	< 0	< 0	< 0	0.32
FEMTO c. 1	7.24	< 0	< 0	< 0	< 0	< 0
FEMTO c. 2	7.04	< 0	< 0	< 0	< 0	< 0

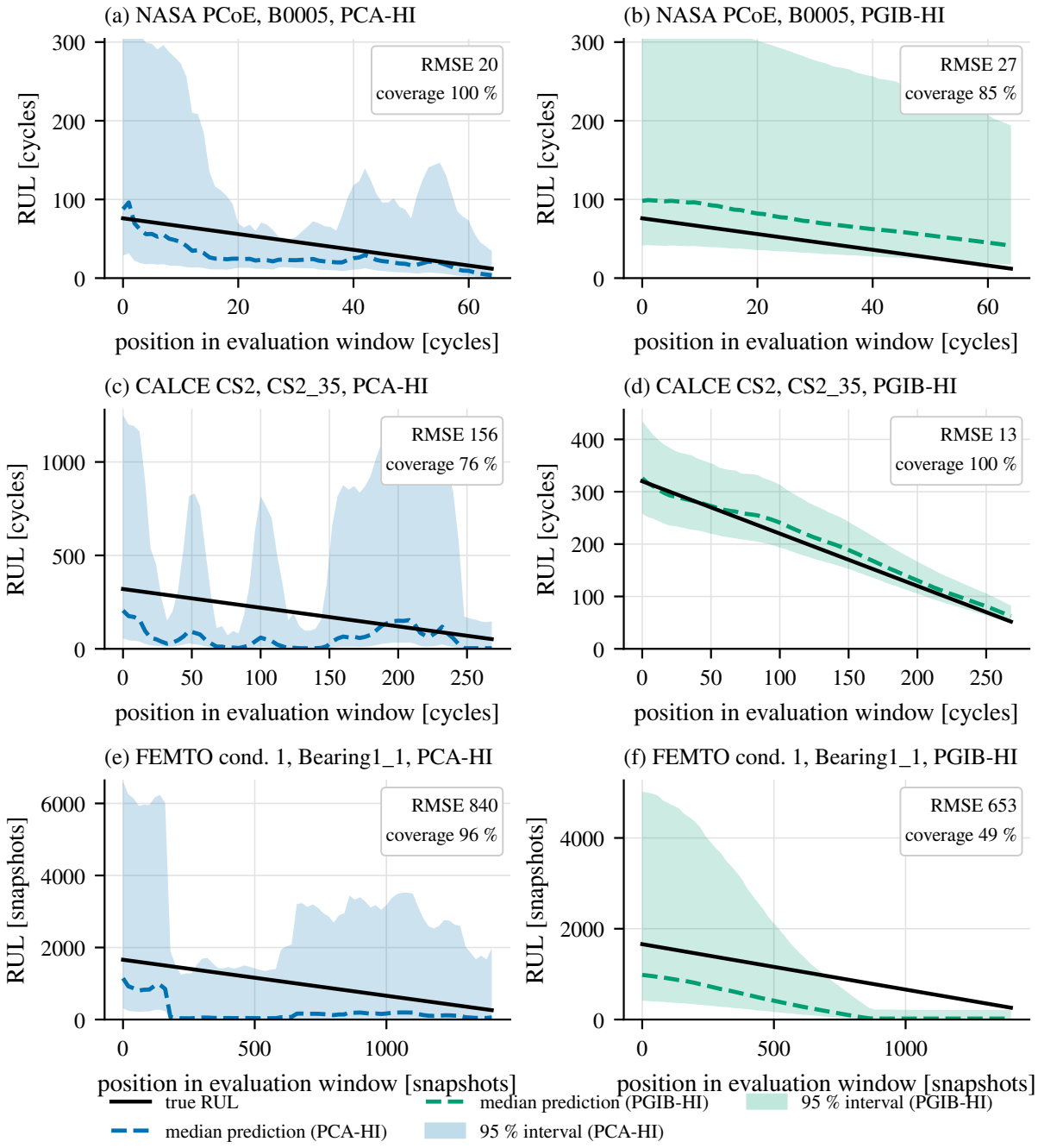


Figure S1: Example RUL predictions with 95% intervals on one held-out unit per data set (NASA B0005, CALCE CS2_35, FEMTO Bearing1_1), for PCA-HI (left) and PGIB (right); each panel gives the RMSE and the coverage on that unit.

(54.7) is only 8%. On bearings the RMSE of the age-only baseline (602 and 664 snapshots for conditions 1 and 2) is comparable to the standard deviation of the pooled RUL, and most HIs do no better: the PGIB has the lowest RMSE in condition 1 (524.5, 13% below the age-only baseline) but gives almost no benefit in condition 2 (621.8 against 664). Two long-lived units dominate the pooled error in condition 2.

The ablations (in the supplied result files) support the bottleneck on all five sets (for example RMSE 54.4 $\rightarrow$ 47.6 on NASA and 42.3 $\rightarrow$ 33.0 on CALCE) and the epistemic drift variance for coverage (for example 14 $\rightarrow$ 42% on NASA). The physics prior helps on

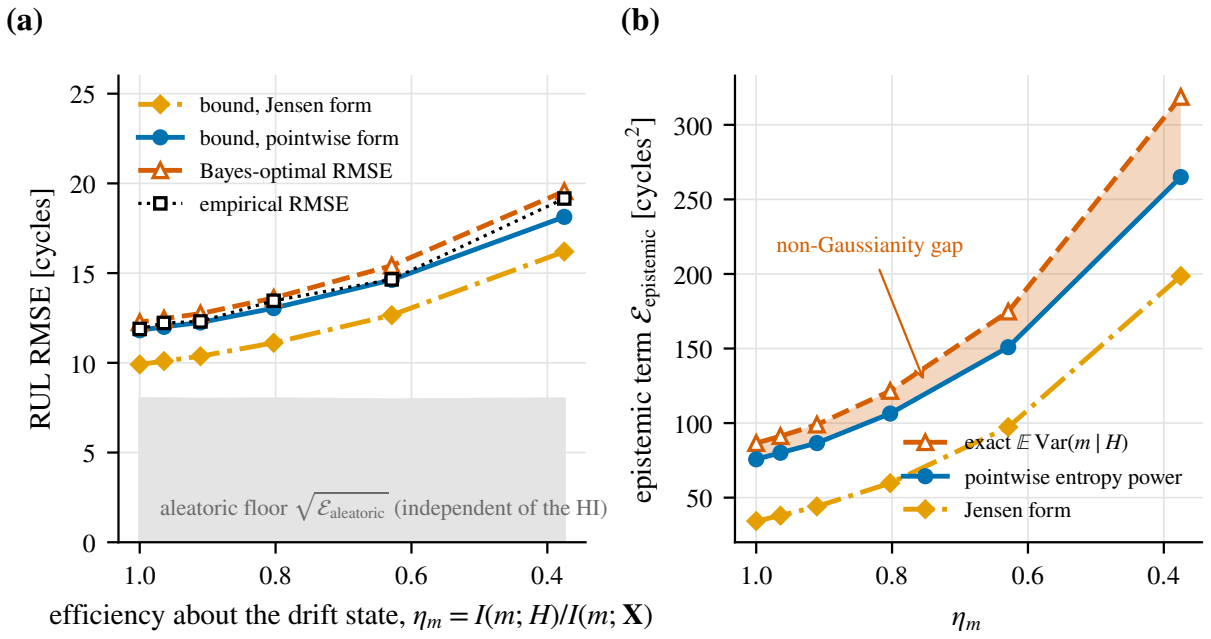


Figure S2: Physics-constrained bound as the HI loses information (decreasing η_m). (a) RMSE of the bounds and estimators. (b) Epistemic term; the shaded area is the non-Gaussianity gap.

CALCE and FEMTO condition 1 but not on NASA or FEMTO condition 2. With only three to six training units per fold, the validation signal that selects the encoder is itself noisy; on NASA it chose the strongest compression ($\beta = 30$), at the edge of the grid, in all eight selections.

Geometric scores versus error. Across the five HIs of each data set, the rank correlation between the geometric score Mon + Tre + Pro and the RMSE is +0.90 on NASA (a higher geometric score goes with a larger error), −1.00 on CALCE, −0.90 on six-cell CALCE, −0.60 on FEMTO condition 1 and +0.60 on condition 2. The geometric criteria therefore did not rank the HIs consistently by prognostic error. This is an association over five methods per data set and not evidence for the mechanism of Theorem 3: the VIB and PGIB HIs have Mon ≈ Tre ≈ 1 because their velocity is constrained to be positive, not because of a squeeze.

S8. Figures and tables moved out of the paper

These figures and tables duplicate numbers that the paper gives in its text and in other tables.

S9. Simulation settings

Original descriptions of the three simulations of Sections 3 and 4.

The drift is $\mu \sim \mathcal{N}(1, 0.22^2)$, truncated to $\mu > 0.3$ and $a > 1$ (which keeps posterior sampling exact), with $\sigma_B = 0.85$, $D_{\text{fail}} = 100$ and $t = 50$. The raw sensor is $X_k = D_k + \xi_k$ with $\xi_k \sim \mathcal{N}(0, \sigma_\xi^2)$ and $\sigma_\xi = 1.5$. HIs with decreasing information are $H_k^\varsigma = X_k + \varsigma\zeta_k$ with independent standard normal ζ_k and $\varsigma \in \{0, 2, 4, 8, 16, 32\}$; each is a degraded version of X. The posterior of (D_t, μ) given H^ς is Gaussian (Kalman filter on (D, μ)), and $m = a/\mu$ is evaluated on 1500 posterior draws for each of 1500 test units; $h(m \mid H)$

Table S16: Template shrinkage of the ridge HI: monotonicity and information lower bound (nats).

Subset		$\gamma = 0.0$	$\gamma = 0.5$	$\gamma = 0.9$	$\gamma = 0.97$	$\gamma = 0.99$	$\gamma = 0.995$	Mon_∞ (predicted)
FD001	Mon	0.24	0.34	0.54	0.56	0.55	0.55	0.55
	lower bound	0.31	0.28	0.21	0.09	-0.06	-0.08	
FD002	Mon	0.19	0.28	0.54	0.58	0.58	0.58	0.61
	lower bound	0.21	0.20	0.18	0.11	0.00	-0.06	
FD003	Mon	0.16	0.22	0.40	0.41	0.42	0.42	0.41
	lower bound	0.10	0.10	0.10	0.07	0.02	-0.07	
FD004	Mon	0.13	0.18	0.36	0.41	0.41	0.41	0.41
	lower bound	0.17	0.18	0.16	0.13	0.07	0.00	
HUST	Mon	0.32	0.53	0.80	0.81	0.81	0.81	0.82
	lower bound	0.30	0.28	0.22	0.12	0.04	-0.00	
MATR	Mon	0.39	0.46	0.50	0.48	0.48	0.48	0.49
	lower bound	0.32	0.28	0.17	-0.02	-0.14	-0.20	
Tongji	Mon	0.65	0.70	0.73	0.73	0.73	0.73	0.70
	lower bound	0.62	0.54	0.33	0.16	0.09	-0.00	

Table S17: Direction of a scalar linear HI: range of stage-optimal angles, best static angle, and worst-stage RMSE ratios to the direction chosen separately at each stage.

Subset	stages	optimal angle [deg]	static angle [deg]	static / free	level only / free	velocity only / free
FD001	life fraction	43–70	70	1.019	1.18	1.07
FD001	age	33–69	60	1.021	1.20	1.18
FD002	life fraction	18–66	55	1.002	1.13	1.05
FD002	age	21–56	45	1.037	1.19	1.29
FD003	life fraction	11–33	20	1.003	1.05	1.16
FD003	age	6–35	25	1.025	1.08	1.38
FD004	life fraction	1–20	5	1.007	1.02	1.10
FD004	age	9–37	15	1.018	1.05	1.33
HUST	life fraction	3–80	70	1.090	1.13	1.10
HUST	age	0–57	20	1.184	1.08	1.54
MATR	life fraction	42–88	65	1.000	1.06	1.01
MATR	age	0–10	0	1.012	1.01	1.42
Tongji	life fraction	57–89	90	1.031	1.35	1.03
Tongji	age	13–55	40	1.068	1.12	1.18

Table S18: Cross-validated age-conditional information lower bounds (nats, 95% bootstrap interval); $\bar{h}(T \mid t)$: mean conditional entropy. Sensors: raw sensors (engines) or charging features (cells); Capacity: reference predictor on the discharge capacity (cells).

Subset	Units	$\bar{h}(T \mid t)$	Age only (control)	Ridge HI	PCA-HI	Level only	Slope only	Sensors	Capacity (ref.)
FD001	100	5.11	-0.11 [-0.28, 0.08]	0.31 [0.20, 0.44]	0.25 [0.13, 0.37]	0.07 [-0.10, 0.24]	0.17 [0.01, 0.32]	0.22 [0.06, 0.38]	–
FD002	260	5.11	-0.11 [-0.22, -0.01]	0.21 [0.13, 0.30]	0.19 [0.11, 0.27]	0.11 [0.03, 0.21]	0.11 [0.01, 0.20]	0.16 [0.08, 0.25]	–
FD003	100	5.64	-0.22 [-0.37, -0.03]	0.10 [-0.05, 0.28]	0.08 [-0.07, 0.25]	0.06 [-0.09, 0.24]	0.00 [-0.13, 0.16]	0.20 [0.04, 0.40]	–
FD004	249	5.52	-0.12 [-0.24, -0.00]	0.18 [0.06, 0.31]	0.02 [-0.11, 0.14]	0.14 [0.01, 0.28]	0.02 [-0.10, 0.14]	0.25 [0.13, 0.38]	–
HUST	77	7.11	-0.02 [-0.14, 0.11]	0.30 [0.19, 0.43]	0.52 [0.42, 0.64]	0.18 [0.06, 0.31]	0.11 [0.01, 0.22]	0.67 [0.57, 0.80]	0.59 [0.48, 0.70]
MATR	169	6.94	-0.27 [-0.46, -0.05]	0.32 [0.17, 0.51]	0.34 [0.17, 0.54]	0.24 [0.07, 0.44]	-0.03 [-0.21, 0.19]	0.61 [0.42, 0.79]	0.29 [0.09, 0.49]
Tongji	88	5.99	0.05 [-0.04, 0.15]	0.60 [0.47, 0.76]	0.52 [0.37, 0.67]	0.33 [0.22, 0.45]	0.27 [0.17, 0.38]	0.91 [0.76, 1.05]	0.40 [0.28, 0.52]

Table S19: RMSE of a linear estimator of the life fraction. Blind: without normalisation by regime (engines) or by the cell's own early cycles (cells).

Subset	Regimes	Blind	Normalised	Regime-specific weights
FD001	1	0.139	0.139	0.139
FD002	6	0.148	0.137	0.136
FD003	1	0.156	0.156	0.156
FD004	6	0.199	0.165	0.164
HUST	one per cell	0.136	0.115	–
MATR	one per cell	0.179	0.142	–
Tongji	one per cell	0.190	0.167	–

Table S20: Age-conditional information lower bounds (nats) of the benchmark methods on the official test engines. †: the age-only control fails.

Method	FD001	FD002	FD003†	FD004
$h(T \mid t)$	5.09	5.07	5.61	5.47
Age-only baseline (control)	0.07 [-0.04, 0.20]	0.01 [-0.07, 0.09]	0.33 [0.23, 0.44]	0.05 [-0.02, 0.13]
PCA-HI	0.59 [0.49, 0.72]	0.29 [0.21, 0.39]	0.67 [0.54, 0.85]	0.26 [0.19, 0.35]
Geo-opt HI	0.58 [0.48, 0.71]	0.23 [0.13, 0.36]	1.10 [0.94, 1.30]	0.66 [0.55, 0.76]
Ridge HI	0.41 [0.26, 0.58]	0.27 [0.17, 0.39]	0.90 [0.77, 1.05]	0.62 [0.54, 0.71]
β-VAE	0.59 [0.47, 0.72]	0.24 [0.16, 0.34]	0.49 [0.35, 0.65]	0.22 [0.14, 0.31]
VIB	0.15 [0.01, 0.32]	0.06 [-0.01, 0.14]	0.57 [0.45, 0.74]	0.11 [0.02, 0.22]
PGIB, integrating (10 seeds)	0.14 [-0.02, 0.32]	0.07 [-0.01, 0.16]	0.49 [0.37, 0.63]	0.19 [0.08, 0.30]
PGIB-KF (10 seeds)	0.61 [0.45, 0.82]	0.28 [0.18, 0.37]	1.04 [0.87, 1.22]	0.48 [0.39, 0.56]
PGIB-KFR (10 seeds)	0.43 [0.28, 0.62]	0.28 [0.20, 0.37]	0.84 [0.73, 0.98]	0.59 [0.51, 0.69]

Table S21: Estimation of η_{info} with the Kraskov–Stögbauer–Grassberger estimator ($k = 5$, 100 replications) in a Gaussian model with known mutual information; n is the sample size.

n	$\hat{I}(T;\mathbf{X})$	$\hat{I}(T;H)$	$\hat{\eta}_{\text{info}}$ (mean $\pm$ standard deviation)	fraction $\hat{\eta}_{\text{info}} > 1$
100	0.23	0.59	2.72 ± 0.74	1.00
300	0.33	0.68	2.10 ± 0.25	1.00
1000	0.43	0.71	1.65 ± 0.09	1.00
3000	0.51	0.71	1.40 ± 0.04	1.00

true values: $I(T;\mathbf{X}) = 1.00$, $I(T;H) = 0.70$, $\eta_{\text{info}} = 0.70$ (nats)

Table S22: Physics-constrained bound. $\mathcal{E}_{\text{ale}}$, $\mathcal{E}_{\text{epi}}$: aleatoric and epistemic terms (cycles^2); I in nats; errors in cycles.

ς	η_m	$I(m;H)$ [nats]	$\mathcal{E}_{\text{ale}}$	$\mathcal{E}_{\text{epi}}$ (exact)	$\mathcal{E}_{\text{epi}}$ (pointwise)	$\mathcal{E}_{\text{epi}}$ (Jensen)	$\sqrt{\text{bound}}$ [cycles]	Bayes RMSE [cycles]
0	1.00	1.40	63.9	86.6	75.7	34.3	11.81	12.26
2	0.96	1.35	64.0	91.2	80.1	37.9	12.00	12.46
4	0.91	1.28	63.4	99.0	86.6	44.1	12.25	12.74
8	0.80	1.13	63.8	121.5	106.4	59.7	13.05	13.61
16	0.63	0.88	62.9	174.7	150.9	97.3	14.62	15.42
32	0.38	0.53	63.8	318.7	265.0	198.6	18.13	19.56

Table S23: Template-shrinkage experiment (mean over three seeds; $\pm$: standard deviation). Information about the velocity V_t in nats: $\mathcal{I}_{\text{state}} = I(V_t;\tilde{H}_t)$, $\mathcal{I}_{\text{kin}} = I(V_t;\Delta\tilde{H}_{2:t} \mid \tilde{H}_t)$, and $\eta_{\text{info}} = I(V_t;\tilde{H}_{1:t})/I(V_t;H_{1:t})$. $\sqrt{\mathcal{B}_{\text{info}}}$: pointwise form, errors in cycles.

γ	Mon	$\mathcal{I}_{\text{state}}$ [nats]	$\mathcal{I}_{\text{kin}}$ [nats]	η_{info}	$\sqrt{\mathcal{B}_{\text{info}}}$ [cycles]	Bayes RMSE [cycles]	Empirical RMSE [cycles]
0.000	0.496	0.789	0.092	1.00	6.67	8.04	15.0±9.8
0.500	0.766	0.779	0.092	0.99	6.81	8.91	14.9±9.4
0.900	0.975	0.567	0.188	0.85	7.38	9.27	15.3±9.2
0.970	0.981	0.159	0.396	0.63	9.01	10.99	17.1±9.2
0.990	0.981	0.022	0.200	0.25	12.87	15.20	21.2±8.9
0.995	0.982	0.006	0.070	0.09	14.88	17.83	23.2±8.9

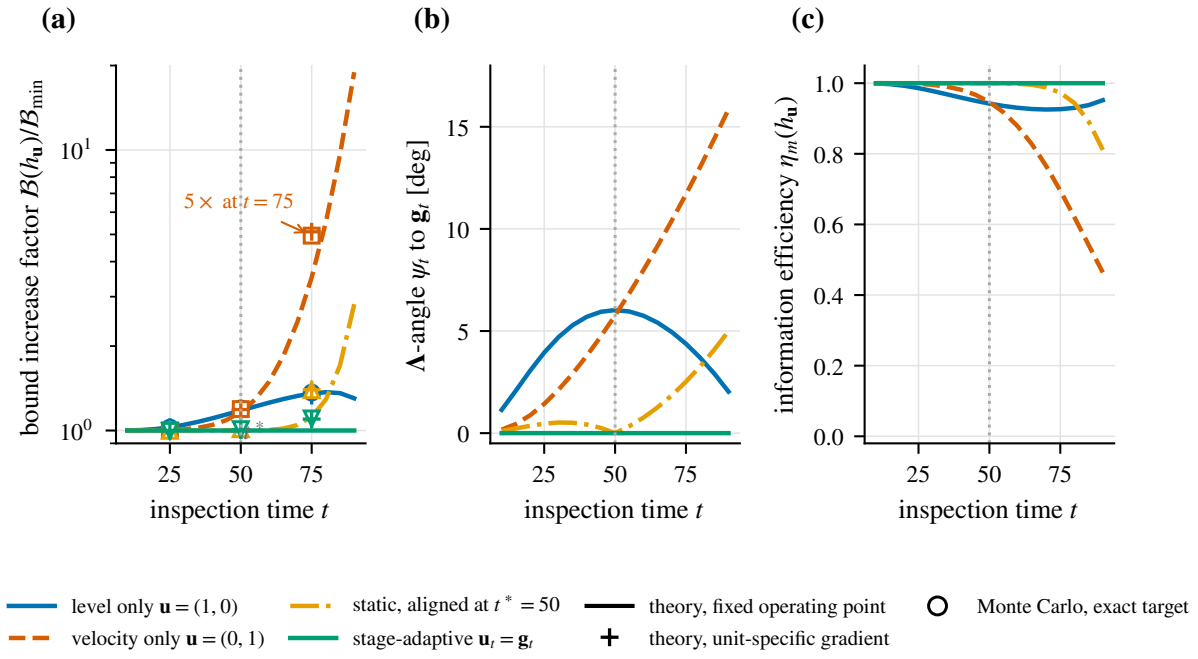


Figure S3: Angular loss relation for scalar linear HIs. (a) Factor by which the bound increases. (b) Angle to the optimal direction $\mathbf{g}_t$ at the fixed operating point. (c) Information efficiency η_m. The static HI is aligned at $t^* = 50$; the stage-adaptive HI uses the stage-wise gradient at the prior mean.

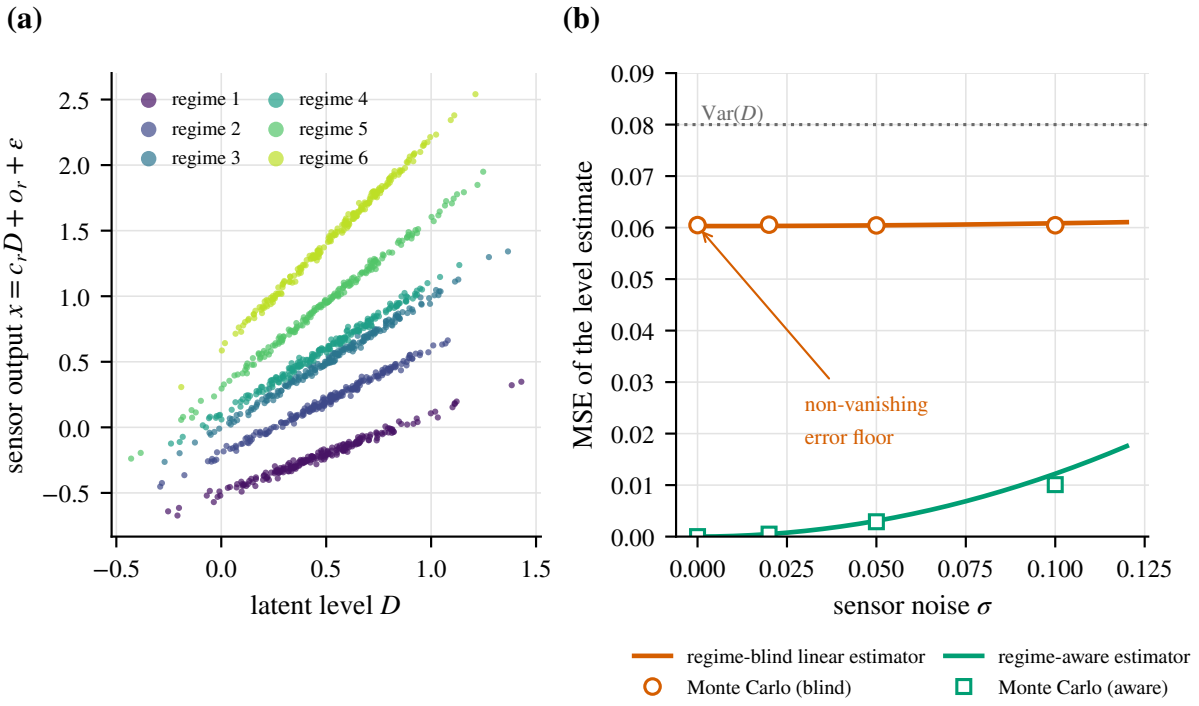


Figure S4: Regime-blind estimation in the six-regime example. (a) Sensor output against latent level for each regime. (b) Error of the regime-blind and regime-aware estimators: formula (lines) and simulation (markers).

and $h(m)$ are estimated with the Vasicek estimator [28]. The Bayes-optimal error is $\mathbb{E}[\mathrm{Var}(T_t \mid \mathbf{z})] + \mathbb{E}[\mathrm{Var}(m \mid H)]$, and the empirical error against simulated first-passage

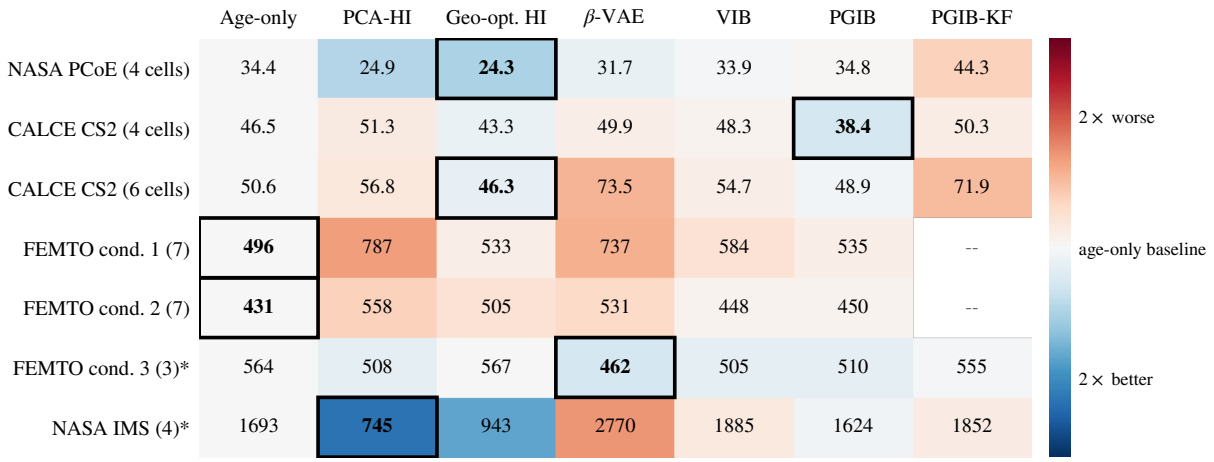


Figure S5: RMSE of the compared HIs on the seven small data sets (cycles for batteries, 10 s snapshots for bearings). Colour: ratio to the age-only baseline; frame: lowest value of each row; cond.: operating condition; number of units in brackets; *: sets not used in the design.

Table S24: Factor by which the bound increases at $t = 50$, mean over 500 units: prediction at the fixed operating point, prediction at the unit's own estimate, and Monte Carlo for the exact target. ψ: angle to the unit-specific gradient.

HI	mean ψ [deg]	fixed point	unit gradient	Monte Carlo
level only $\mathbf{u} = (1, 0)$	6.0	1.179	1.177	1.181
velocity only $\mathbf{u} = (0, 1)$	5.8	1.164	1.194	1.197
stage-fixed gradient (prior mean)	0.7	1.000	1.004	1.003
unit-specific gradient	0.0	1.004	1.000	0.999

times is also reported. Errors are given as RMSE.

The latent state is a local linear trend, $D_k = D_{k-1} + V_{k-1} + \sigma_B w_k$ and $V_k = V_{k-1} + \sigma_V \zeta_k$, with level D_k, velocity V_k, independent standard normal noises w_k, ζ_k and $V_0 \sim \mathcal{N}(\bar{v}, \sigma_{v0}^2)$. The raw HI is $H_k = D_k + \xi_k^{(1)}$ with measurement noise $\xi_k^{(1)} \sim \mathcal{N}(0, \sigma_1^2)$, and the shrinkage is that of Theorem 3(b) with noise standard deviation σ_2. The parameters are $\bar{v} = 2$, $\sigma_{v0} = 0.25$, $\sigma_V = 0.03$, $\sigma_B = 0.5$, $\sigma_1 = 2$, $\sigma_2 = 0.6$, failure threshold $D_{\text{fail}} = 200$ and inspection at $t = 50$. The information about V_t is exact from the Kalman covariances [29, 30]. For the RUL, the Bayes-optimal estimator is the posterior mean from 300 exact posterior draws per unit, $h(T_t \mid \tilde{H})$ is estimated with the Vasicek estimator [28], and $\sqrt{\mathcal{B}_{\text{info}}}$ is the square root of the average of the pointwise entropy powers. Errors are reported as RMSE; each setting uses 400 test units and three seeds.

The model is the local linear trend of Section 4.1 with the raw sensor and $D_{\text{fail}} = 200$. The posterior of $\mathbf{z} = (D_t, V_t)$ is Gaussian, and $\mathbf{\Lambda}$ is strongly anisotropic (eigenvalues

Table S25: C-MAPSS, ten seeds per learned encoder. Single: RMSE, mean ± standard deviation over seeds; ens. 10: ten-seed ensemble; the best simple HI includes the regime-specific ridge regression; last column: paired difference of PGIB-KF from the best simple HI with 95% bootstrap interval over engines [27].

Subset	Best simple HI (RMSE)	PGIB-HI single	PGIB-KF-HI single	PGIB-KFR-HI single	PGIB-KF-HI ens. 10	PGIB-KFR-HI ens. 10	KF − best paired, ens. 10 [95% CI]	KFR − best
FD001	PCA (21.8)	34.8±2.9	23.8±3.3	27.5±3.2	21.3	25.5	-0.6 [-3.2, +2.5]	+3.6 [+0.2, +7.0]
FD002	regime ridge (28.4)	36.3±1.5	31.0±2.2	30.6±2.2	29.2	29.1	+0.8 [-0.7, +2.3]	+0.7 [-1.2, +2.8]
FD003	Geo-opt. (21.9)	41.2±1.9	30.1±4.6	33.8±6.5	23.3	28.4	+1.4 [-1.6, +4.5]	+6.5 [+2.4, +10.6]
FD004	Geo-opt. (29.7)	48.7±2.5	40.7±7.6	36.8±4.0	35.8	31.8	+6.1 [+2.5, +9.7]	+2.1 [-0.9, +5.0]

0.004 and 204). For each of 500 units, the exact entropy power of the *nonlinear* target $m = (D_{\text{fail}} - D_t)/V_t$ given $h_{\mathbf{u}}$ is estimated from the exact conditional Gaussian of $\mathbf{z}$ (2000 draws, Vasicek estimator) and compared with $1 + \rho \sin^2 \psi$ computed in two ways: at a fixed operating point, the prior mean of the stage, as in Theorem 4; and at the unit's own posterior mean, a local analysis in the sense of Remark 2. As a consistency check, in the constant-drift model of Section 3 $\mathbf{\Lambda}$ is effectively of rank one and the predicted factor is 1.000 for every $\mathbf{u}$ (Monte Carlo: 1.000 to 1.003), as the remark after Theorem 2 implies.

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